

CSE 4125: Distributed Database Systems Chapter – 5

Translation of Global Queries to
Fragment Queries.
(Part – E)

Topics to be discussed –

- Simplification of Non-distributed Query

Question type:

/* A query is given */

Now, answer the following questions.

- i. Draw the *operator tree*. [2]
- ii. Perform step-by-step transformations to simplify the operator tree, indicating which rule and criterion is applied at each step. [5]

Non-distributed (Equivalence Transformation for Queries):

- ❑ Query $\rightarrow$ Operator Tree.
- ❑ Operator Tree $\rightarrow$ Simplified Operator Tree.

We will follow this -

****This serial can be changed**

Equivalence Query transformation steps:

1. **Generate the equivalent operator tree (T_{global}) for the given query (Q_{global}).**
2. **Find the common sub-expression (R) from T_{global} .**
3. **Apply rules to remove R and obtain simplified tree T_{removed} .**
4. **Apply criteria – 1 and 2 on T_{removed} to obtain final simplified operator tree $T_{\text{transformed}}$.**
5. **Write the query $Q_{\text{transformed}}$ from $T_{\text{transformed}}$.**

$$\text{So, } Q_{\text{global}} \leftrightarrow Q_{\text{transformed}}$$

Some Rules/Properties

Properties

- $R \text{ NJN } R \leftrightarrow R$  1
- $R \text{ UN } R \leftrightarrow R$  2
- $R \text{ DF } R \leftrightarrow 0$  3
- $R \text{ NJN } \text{SL}_F R \leftrightarrow \text{SL}_F R$  4
- $R \text{ UN } \text{SL}_F R \leftrightarrow R$  5
- $R \text{ DF } \text{SL}_F R \leftrightarrow \text{SL}_{\text{NOT } F} R$  6
- $(\text{SL}_{F1} R) \text{ NJN } (\text{SL}_{F2} R) \leftrightarrow \text{SL}_{F1 \text{ AND } F2} R$  7
- $(\text{SL}_{F1} R) \text{ UN } (\text{SL}_{F2} R) \leftrightarrow \text{SL}_{F1 \text{ OR } F2} R$  8
- $(\text{SL}_{F1} R) \text{ DF } (\text{SL}_{F2} R) \leftrightarrow \text{SL}_{F1 \text{ AND NOT } F2} R$  9

They will be used to remove common sub-expressions in the simplification of operator tree.

Example 1

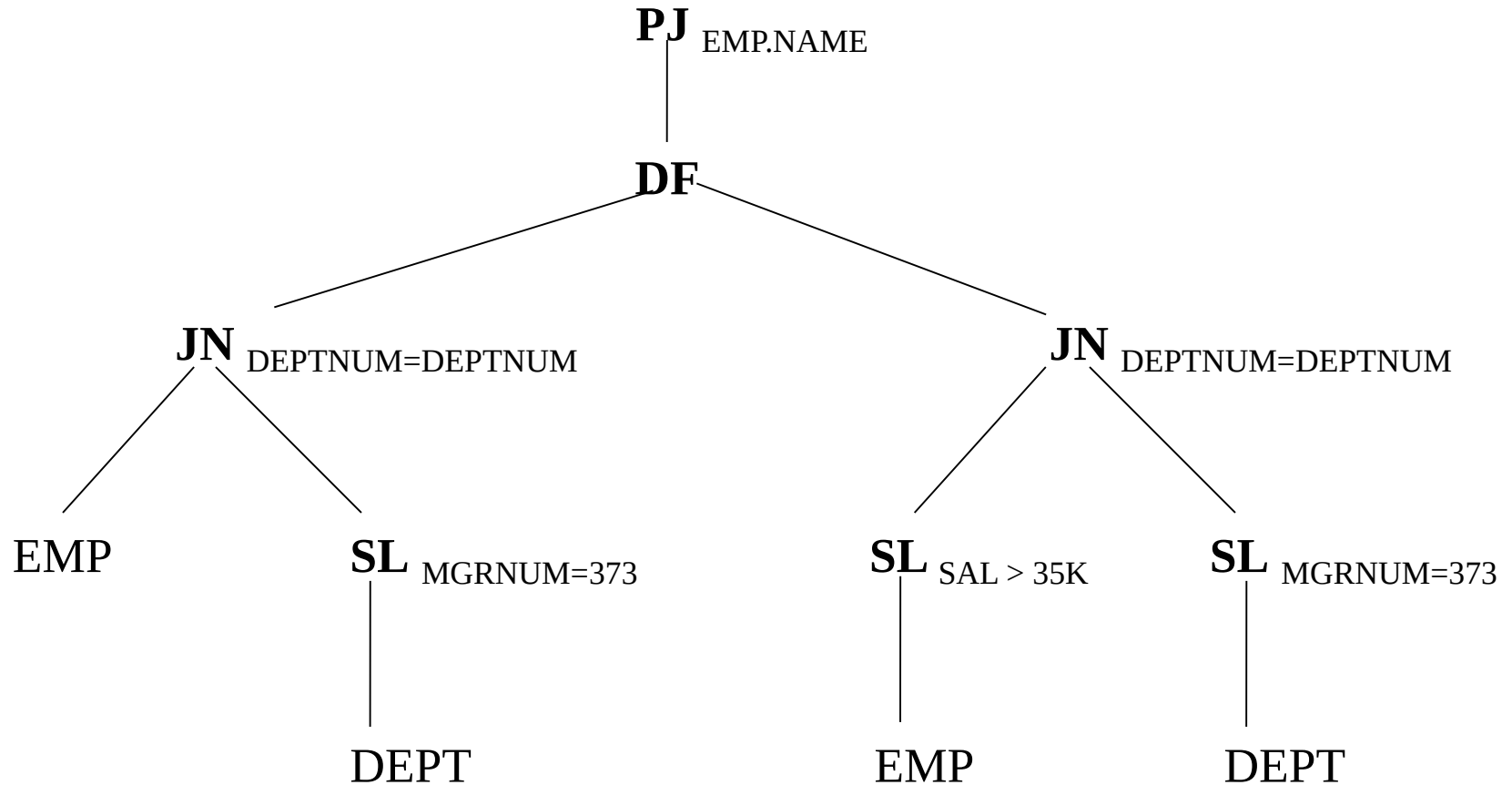
EMP (EMPNUM, DEPTNUM, NAME, SAL, AGE)
DEPT (DEPTNUM, NAME, AREA, MGRNUM)

Q: PJ *EMP.NAME ((EMP JN DEPTNUM=DEPTNUM SL MGRNUM=373 DEPT) DF (SL SAL > 35K EMP JN DEPTNUM=DEPTNUM SL MGRNUM=373 DEPT))*

Now, answer the following questions.

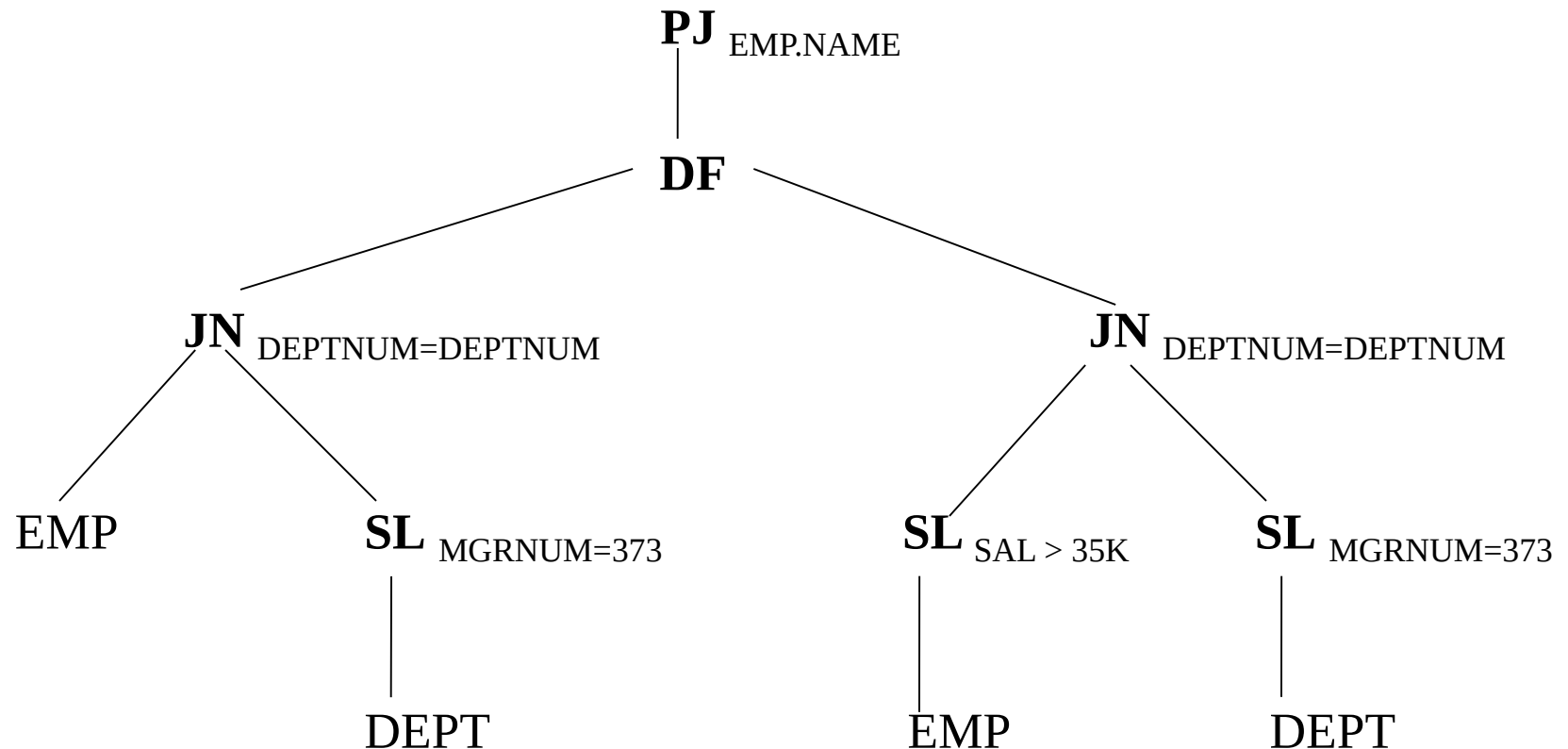
- i. Draw the *operator tree*. [2]
- ii. Perform step-by-step transformations to simplify the operator tree, indicating which rule and criterion is applied at each step. [5]

Operator Tree



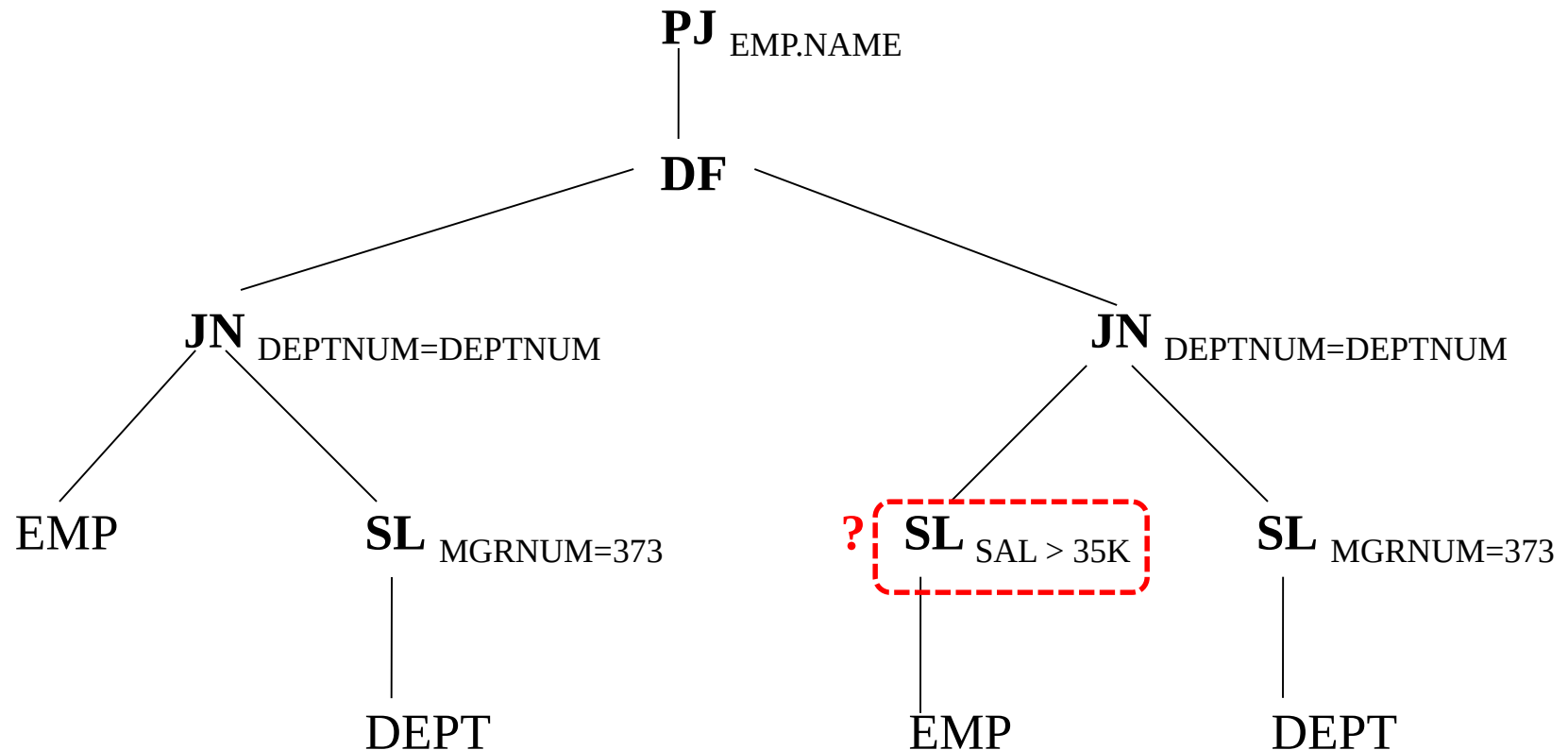
Finding Common Sub-expression

Any common portion?



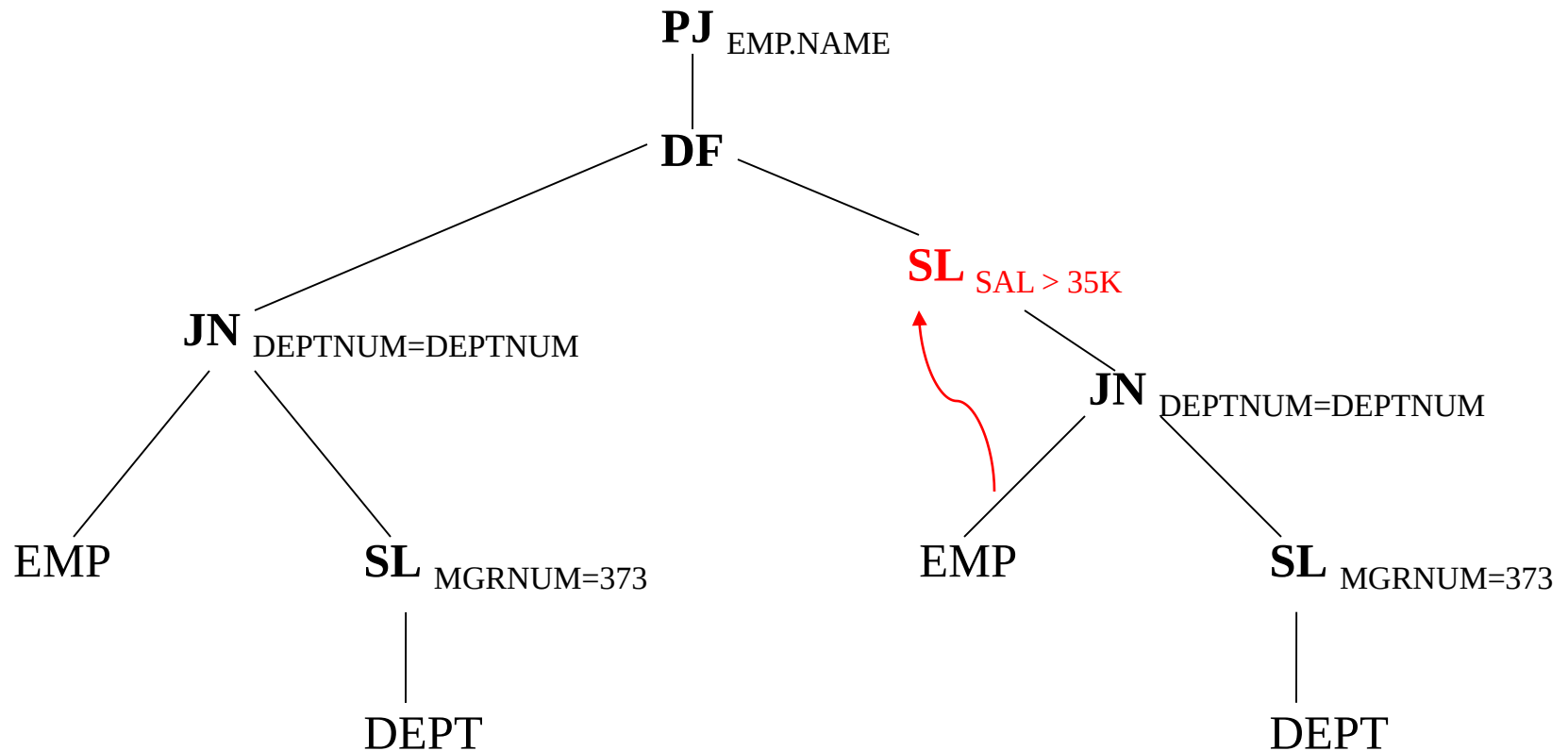
Finding Common Sub-expression

Any common portion?

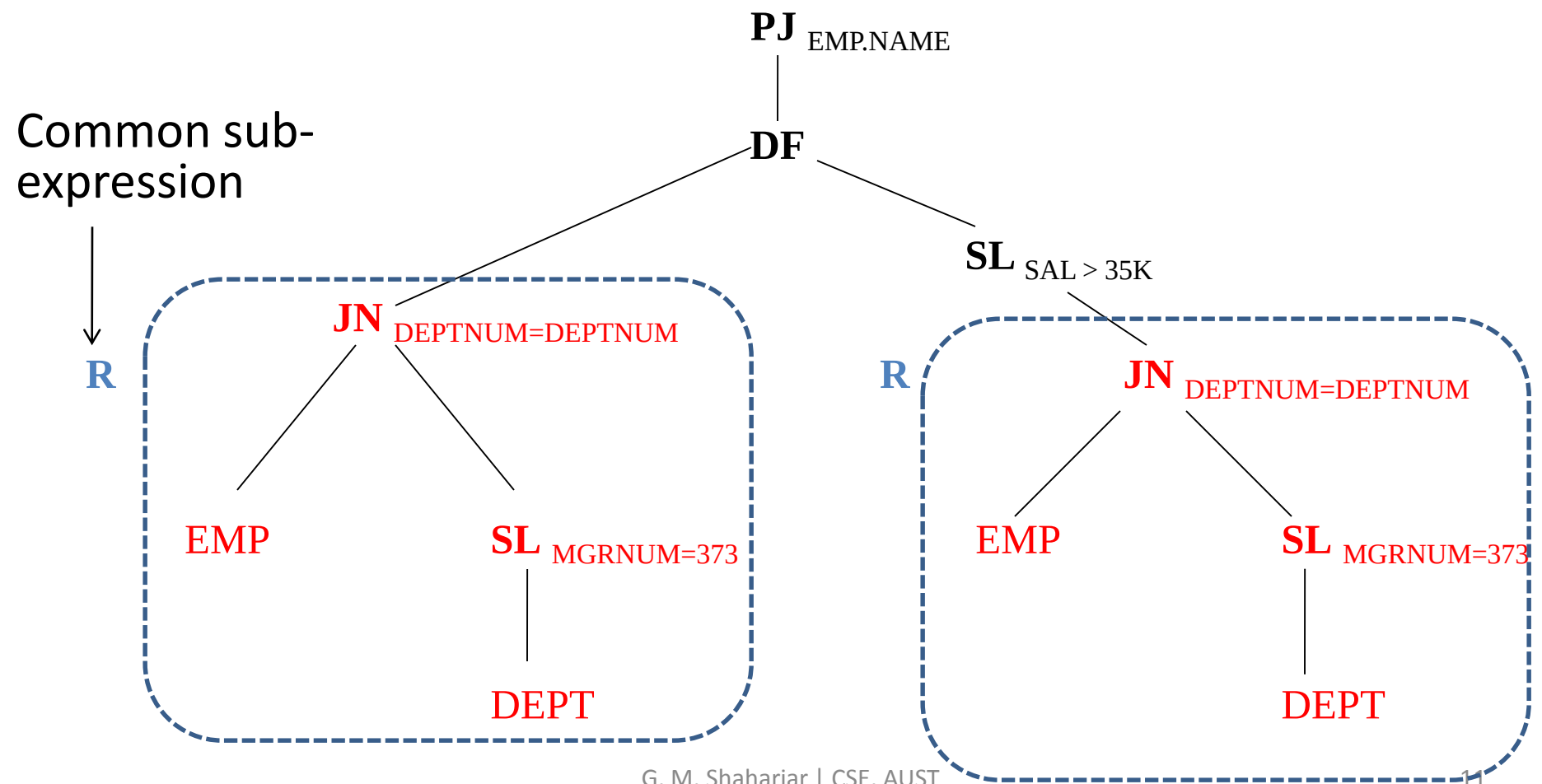


Finding Common Sub-expression

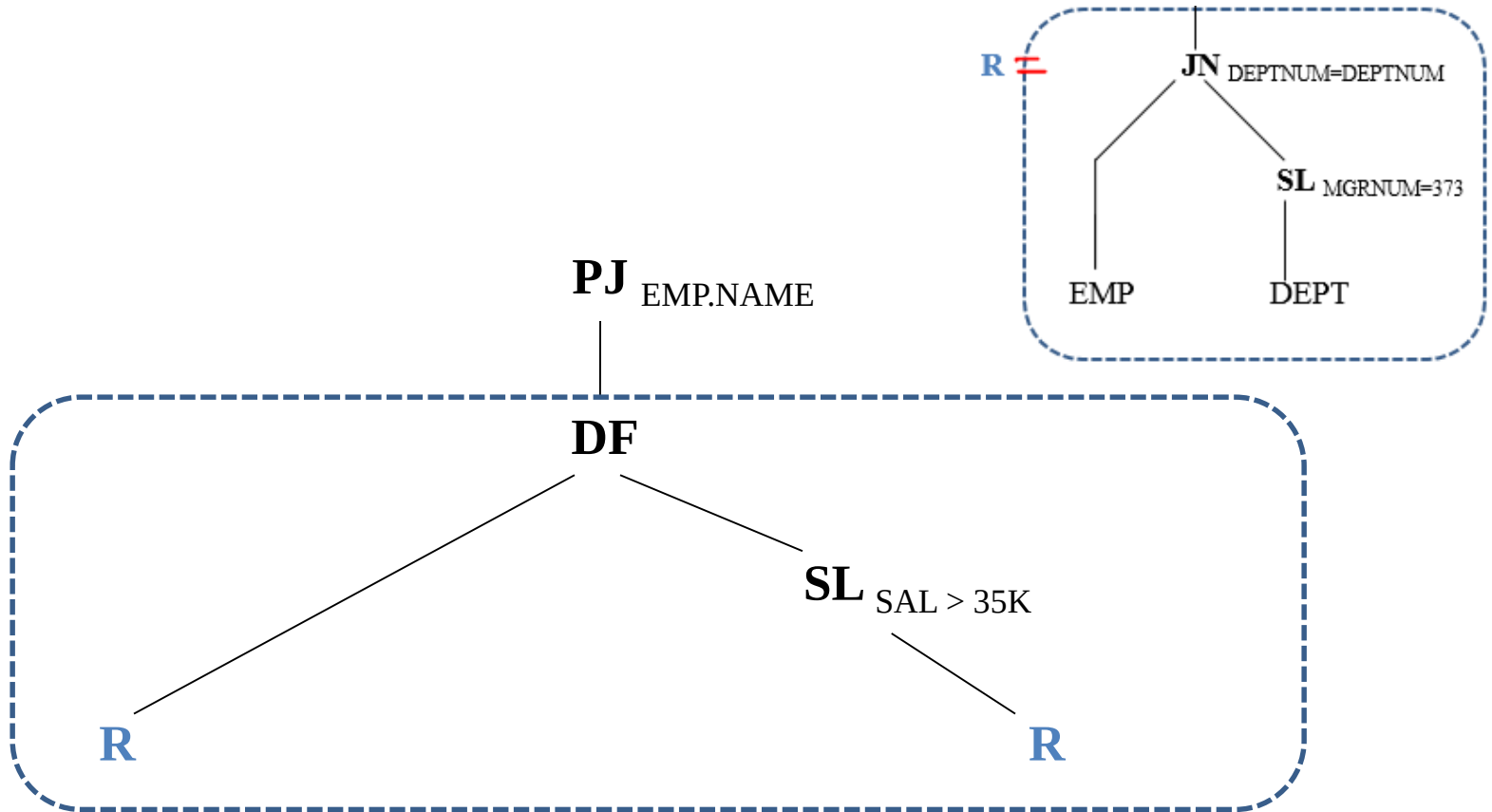
Any common portion? NOW?



Finding Common Sub-expression

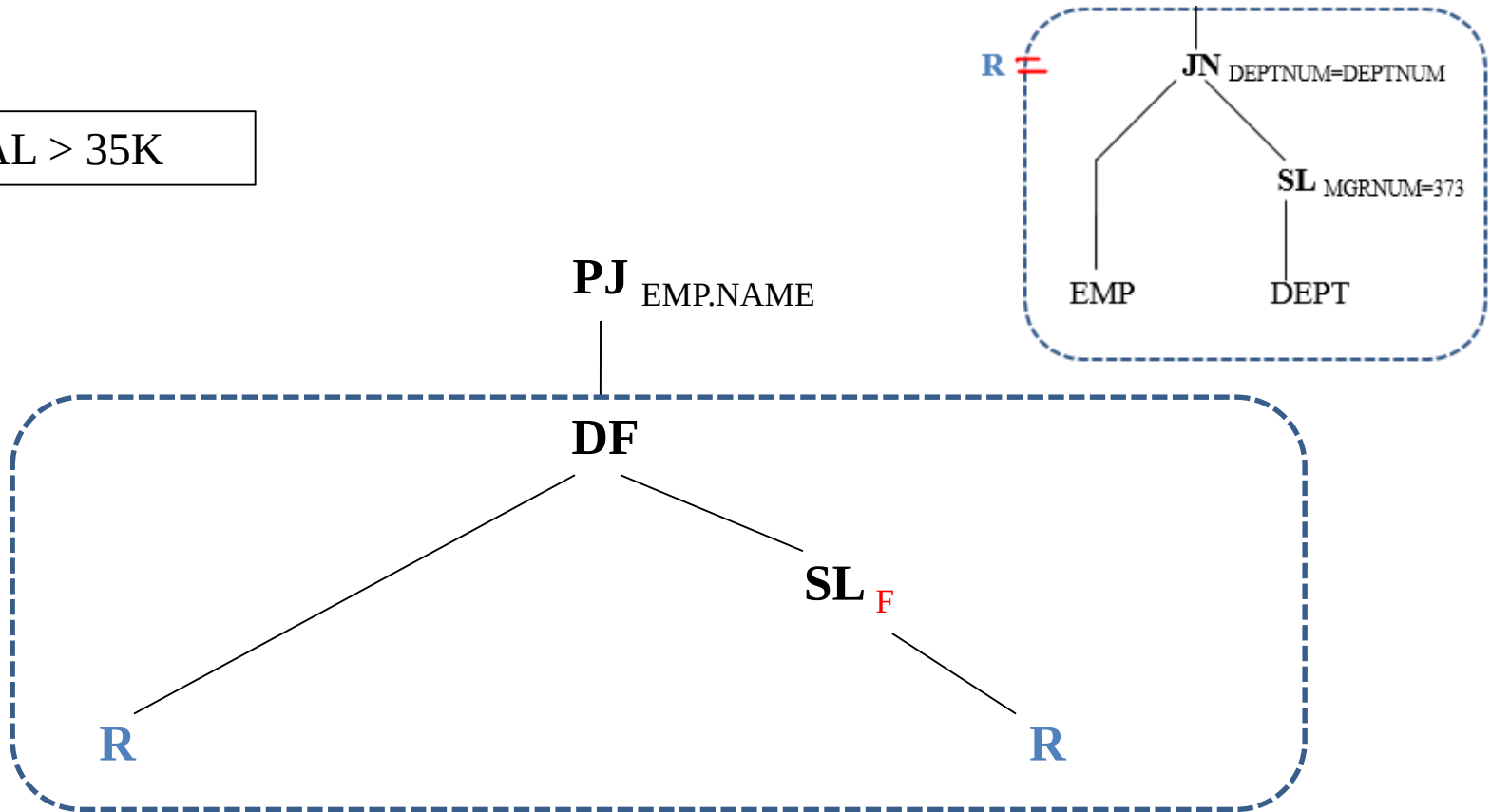


Finding Common Sub-expression



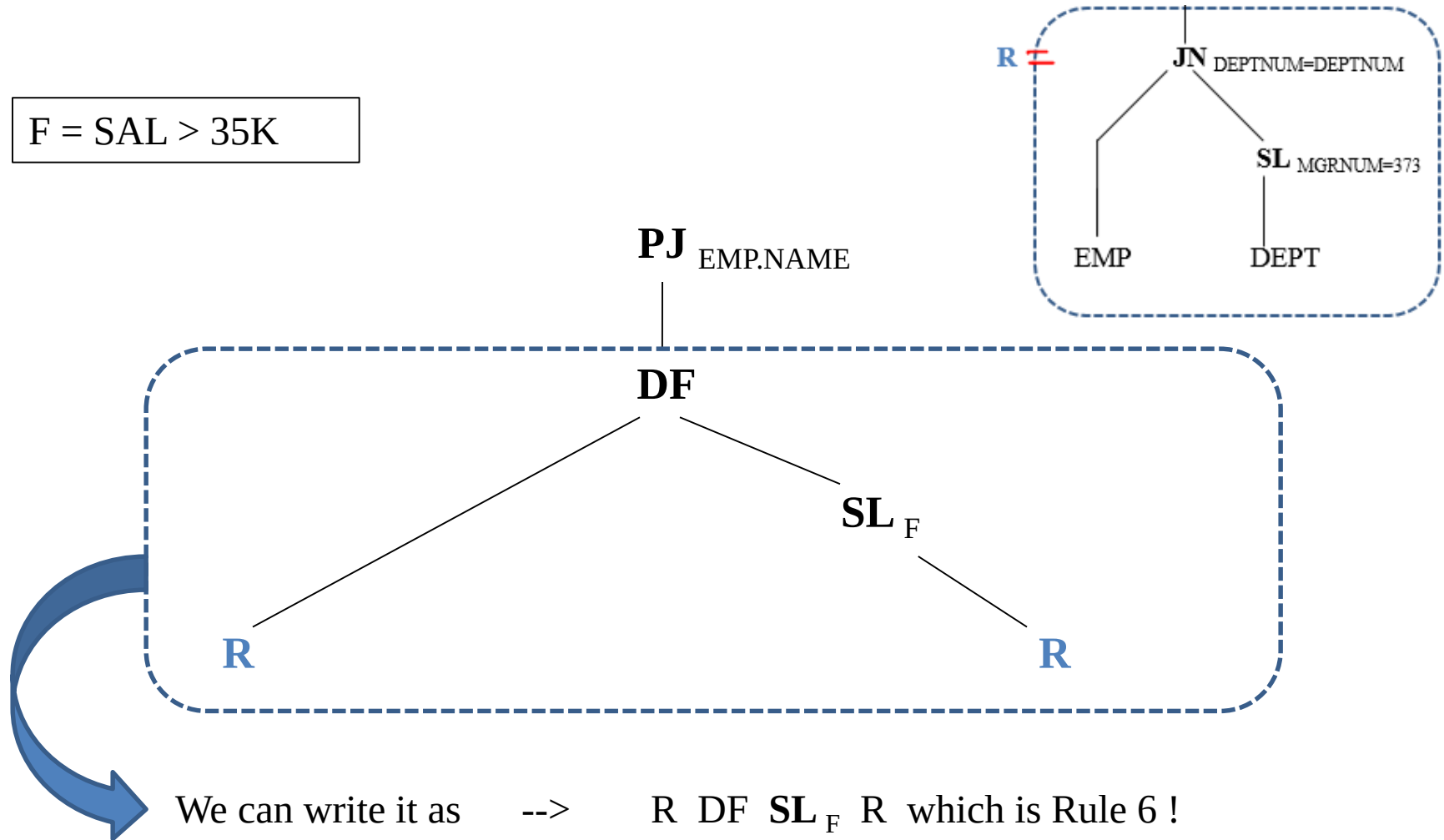
Finding Common Sub-expression

$F = \text{SAL} > 35\text{K}$



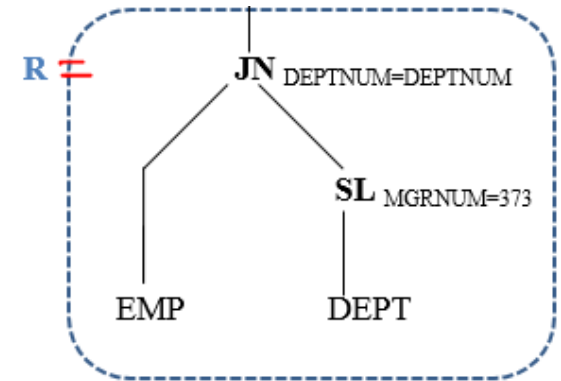
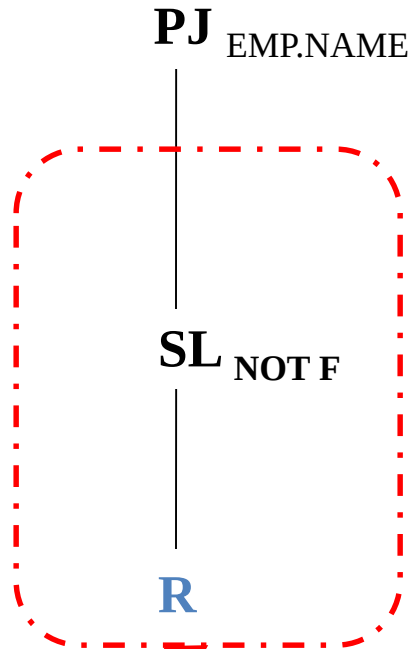
Finding Common Sub-expression

$F = \text{SAL} > 35\text{K}$



Removing Common Sub-expression

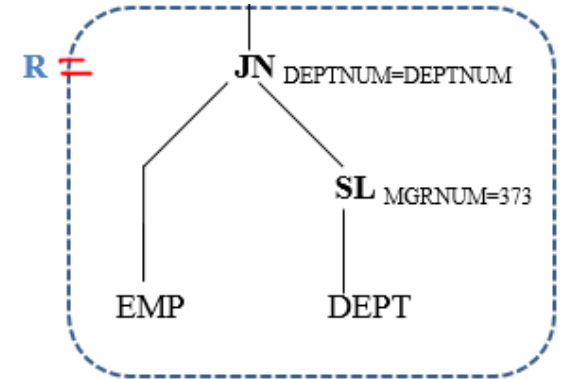
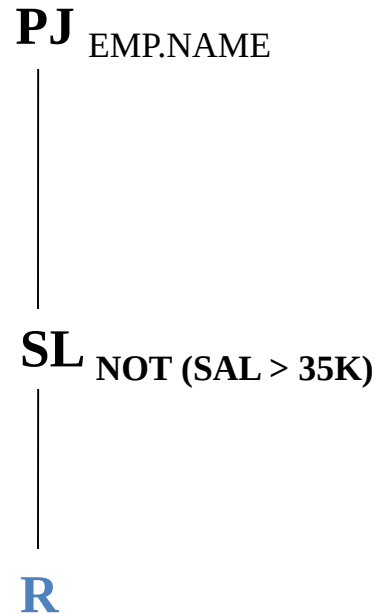
$F = \text{SAL} > 35K$



• $R \text{ DF } SL_F R \leftrightarrow SL_{NOT F} R$

Removing Common Sub-expression

$F = \text{SAL} > 35\text{K}$

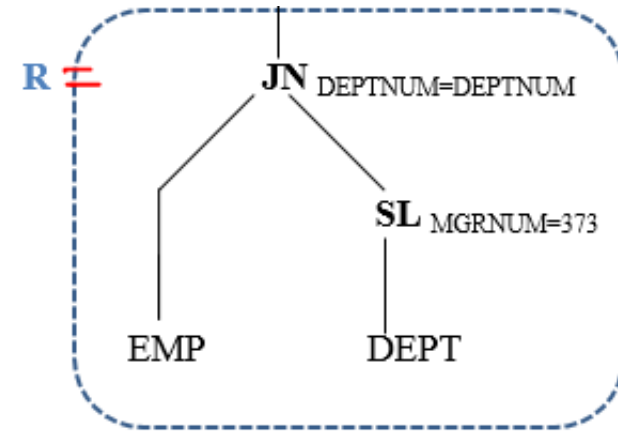


Removing Common Sub-expression

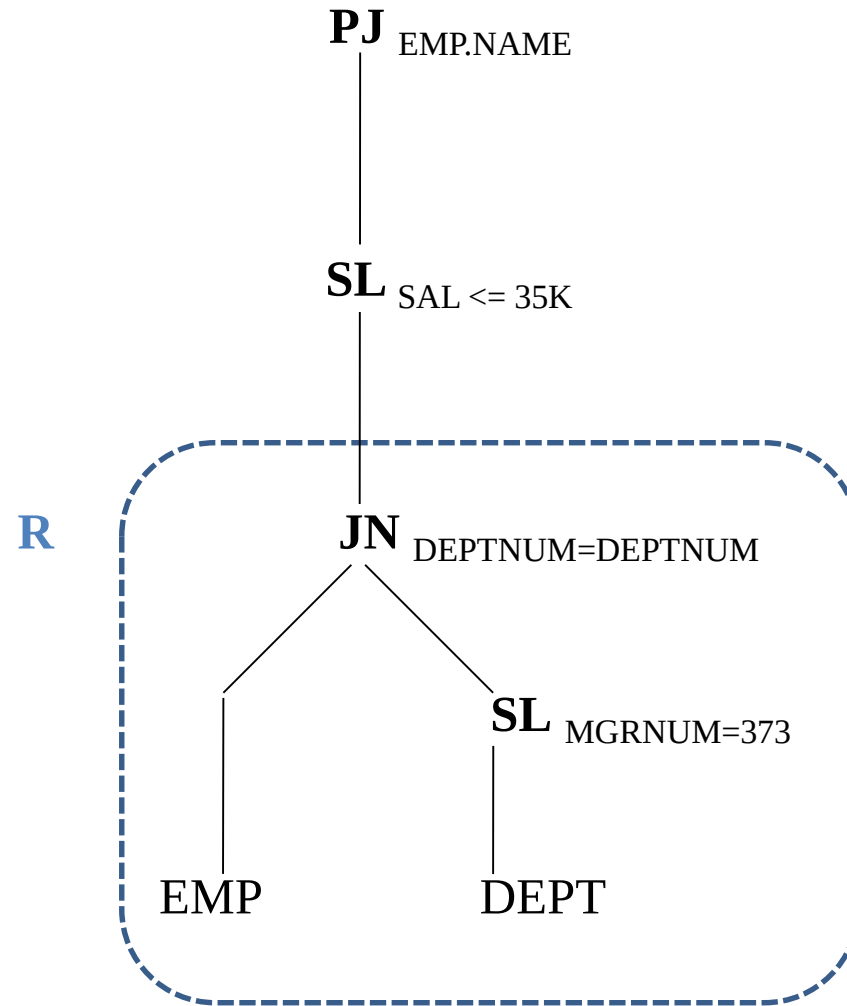
PJ EMP.NAME

SL SAL <= 35K

R



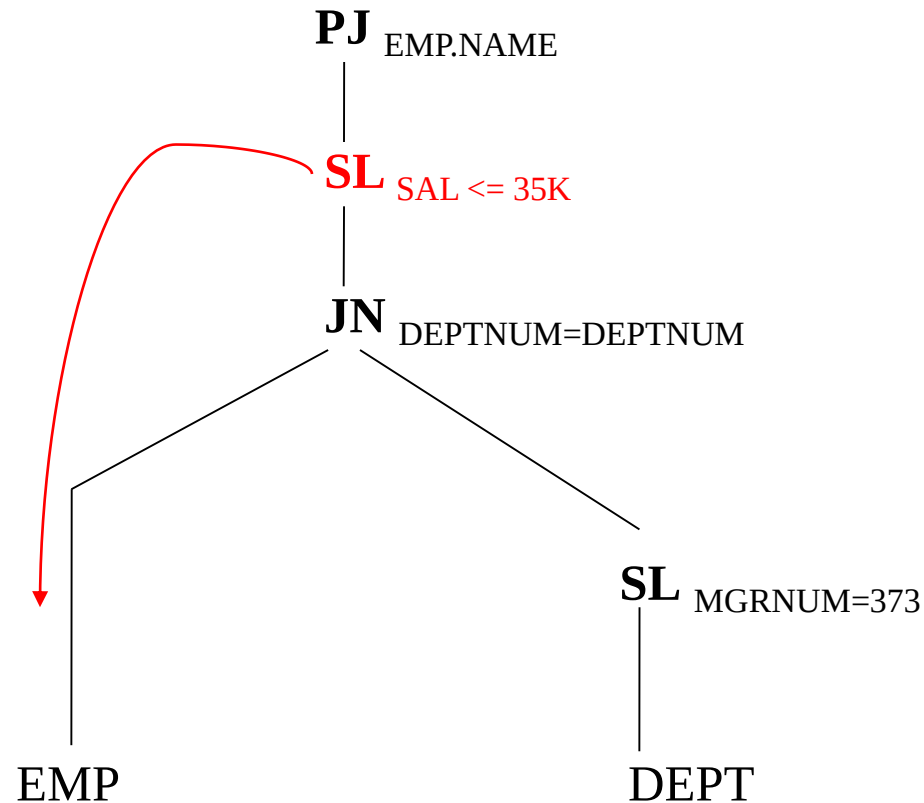
Removing Common Sub-expression



Can you apply Criterion 1 and/or 2 on this tree?

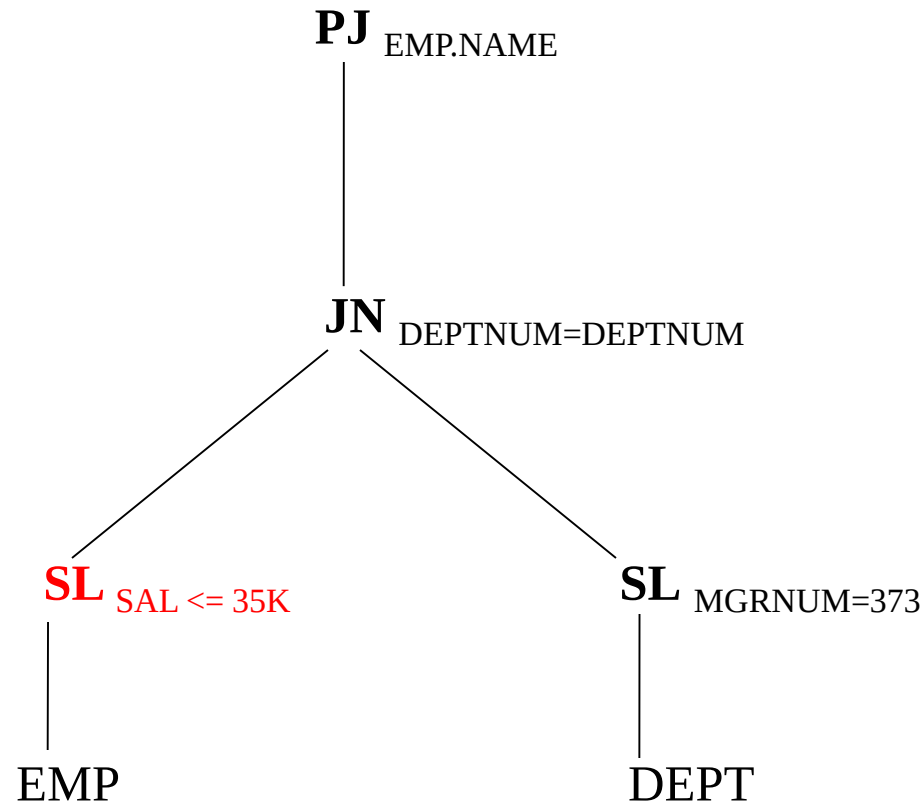
Simplification

Applying criterion – 2



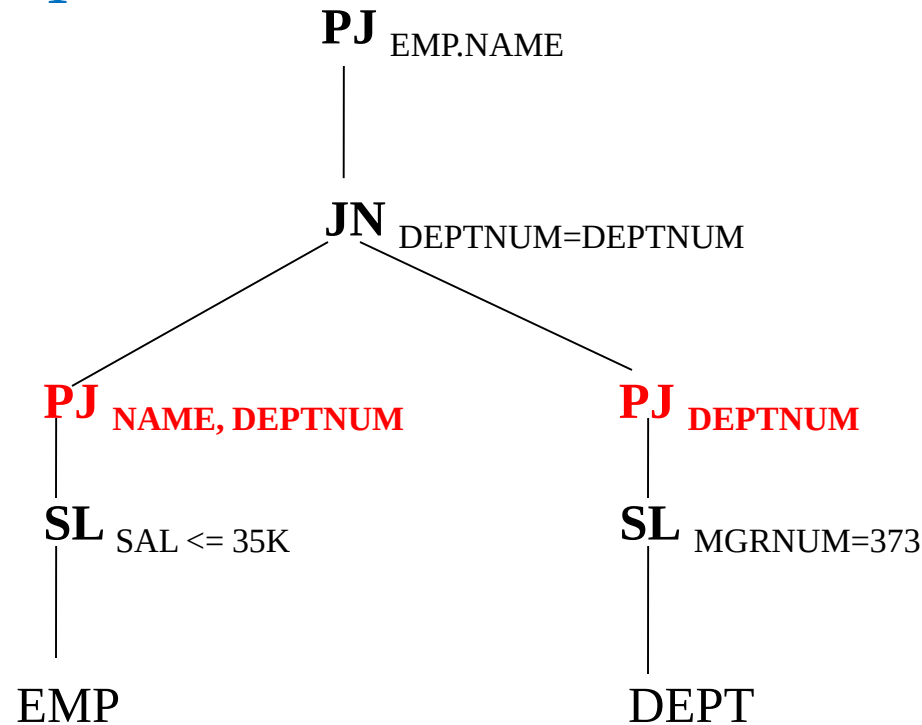
Simplification

After Applying criterion – 2



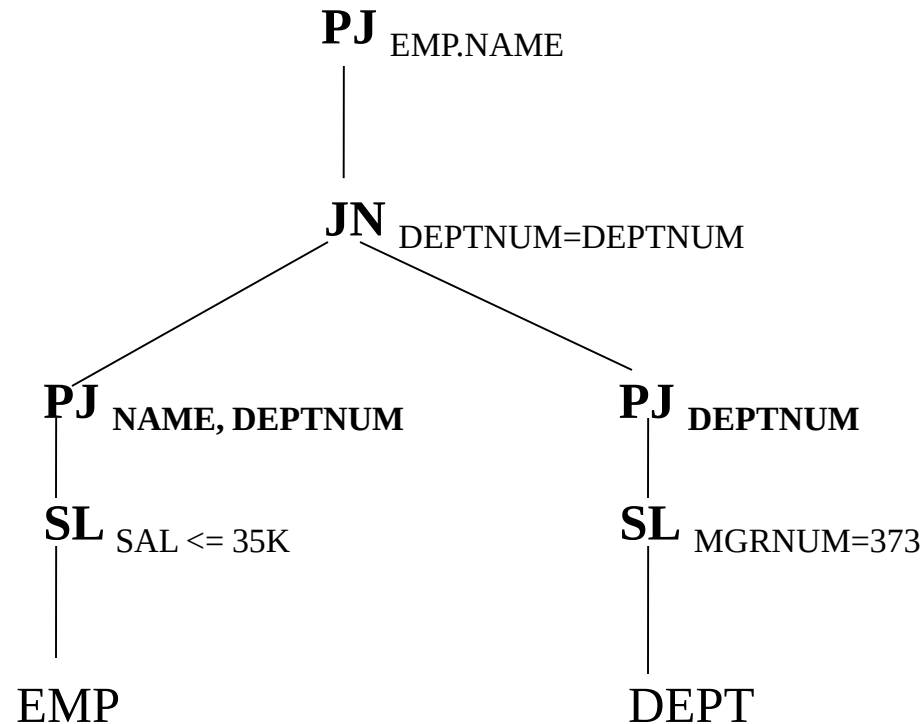
Simplification

After Applying criterion – 1



Transformed Query

Q_T: PJ_{EMP.NAME} ((PJ_{NAME,DEPTNUM} SL_{SAL<=35K} EMP) JN_{DEPTNUM=DEPTNUM} (PJ_{DEPTNUM} SL_{MGRNUM=373} DEPT))



Transformed Query

Output:

Q_T: **PJ**_{EMP.NAME} ((**PJ**_{NAME,DEPTNUM} **SL**_{SAL<=35K} *EMP*) **JN**_{DEPTNUM=DEPTNUM}
(**PJ**_{DEPTNUM} **SL**_{MGRNUM=373} *DEPT*))

Input:

Q: **PJ**_{EMP.NAME} ((*EMP* **JN**_{DEPTNUM=DEPTNUM} **SL**_{MGRNUM=373} *DEPT*)
DF (**SL**_{SAL > 35K} *EMP* **JN**_{DEPTNUM=DEPTNUM} **SL**_{MGRNUM=373} *DEPT*))

$$Q \longleftrightarrow Q_T$$

Example 2.1

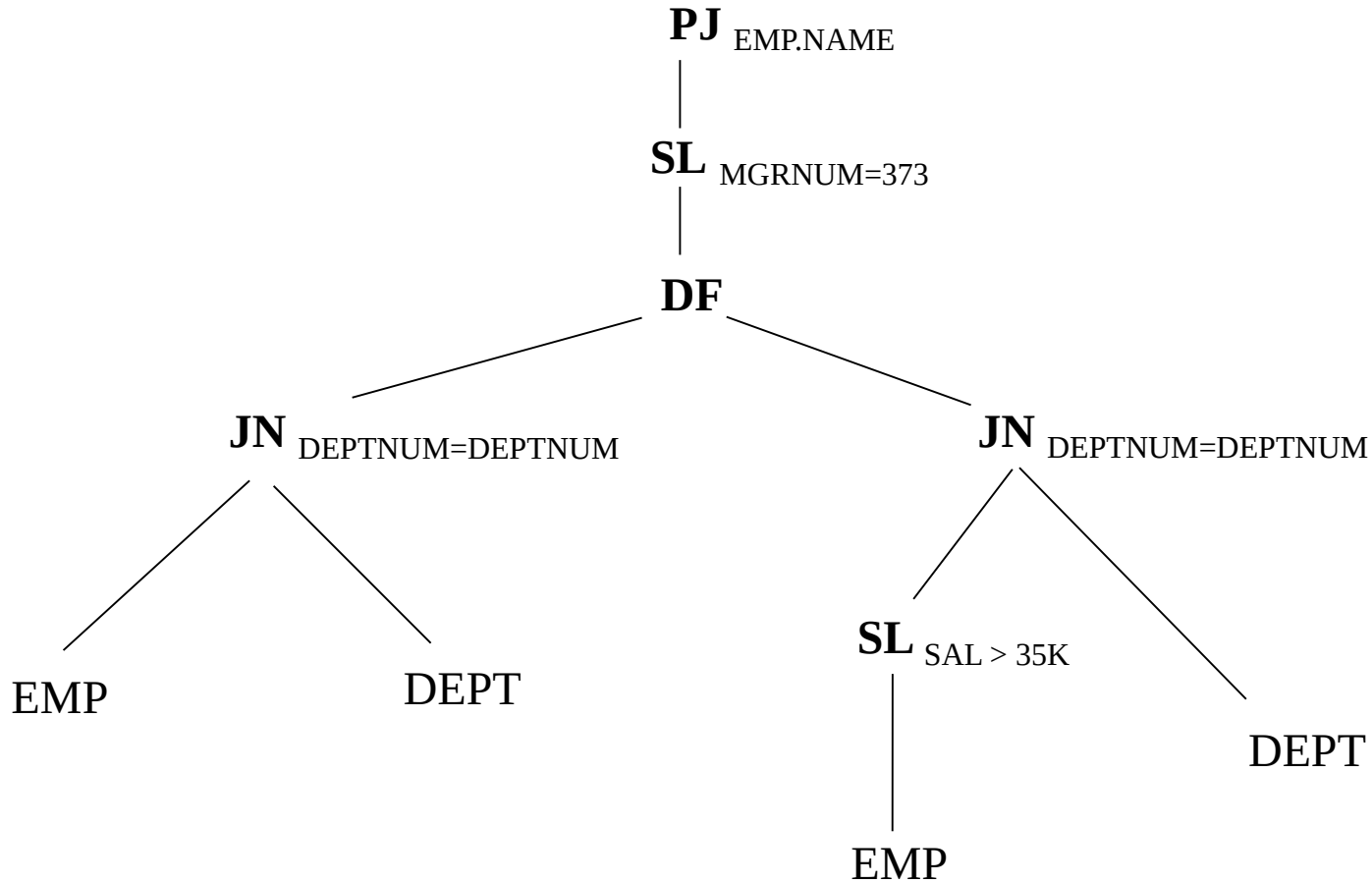
EMP (EMPNUM, DEPTNUM, NAME, SAL, AGE)
DEPT (DEPTNUM, NAME, AREA, MGRNUM)

Q: PJ *EMP.NAME* **SL** *MGRNUM=373* ((*EMP JN* *DEPTNUM=DEPTNUM* *DEPT*) **DF** (*SL* *SAL > 35K* *EMP JN* *DEPTNUM=DEPTNUM* *DEPT*))

Now, answer the following questions.

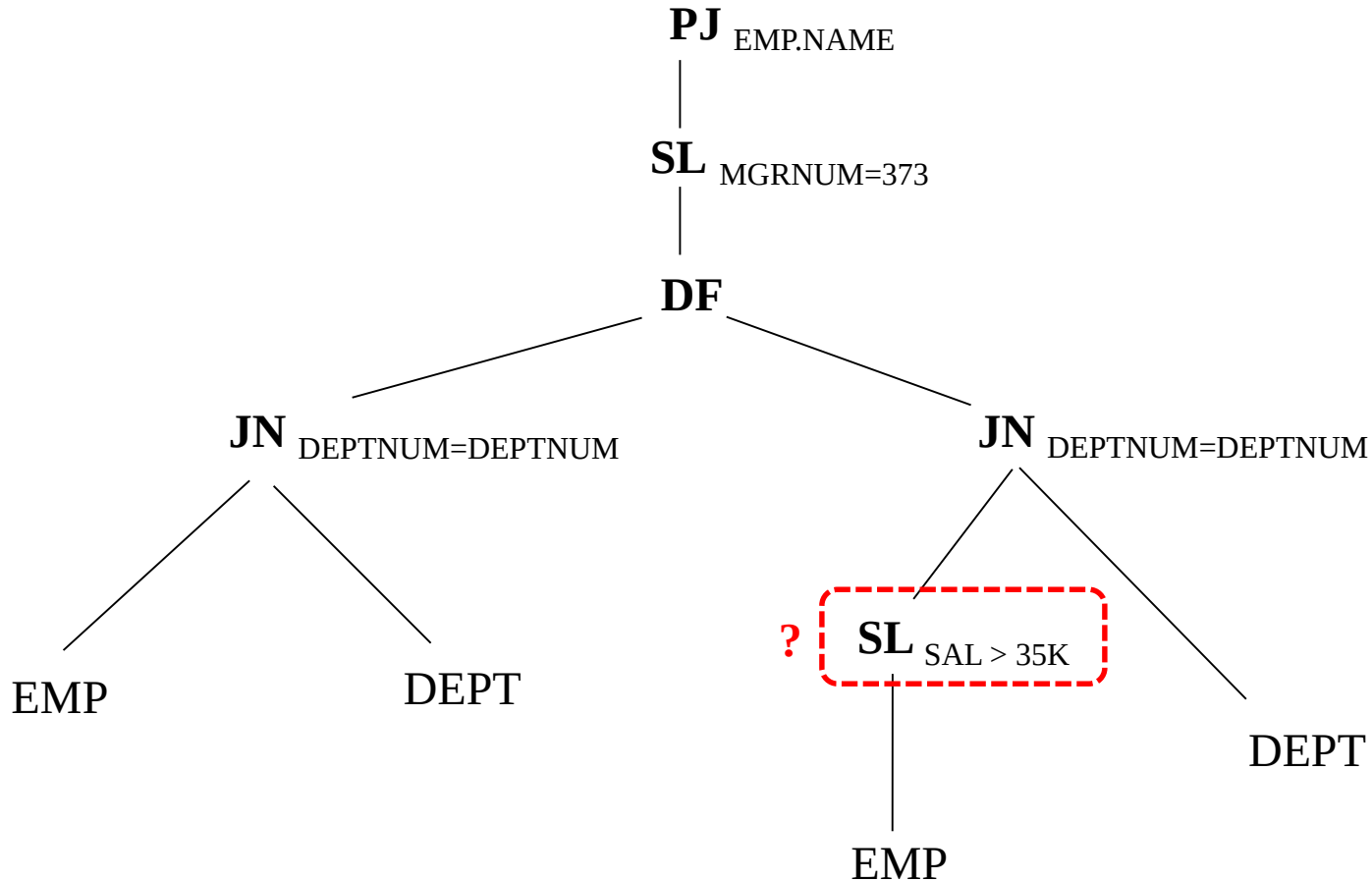
- i. Draw the *operator tree*. [2]
- ii. Perform step-by-step transformations to simplify the operator tree, indicating which rule and criterion is applied at each step. [5]

Operator Tree



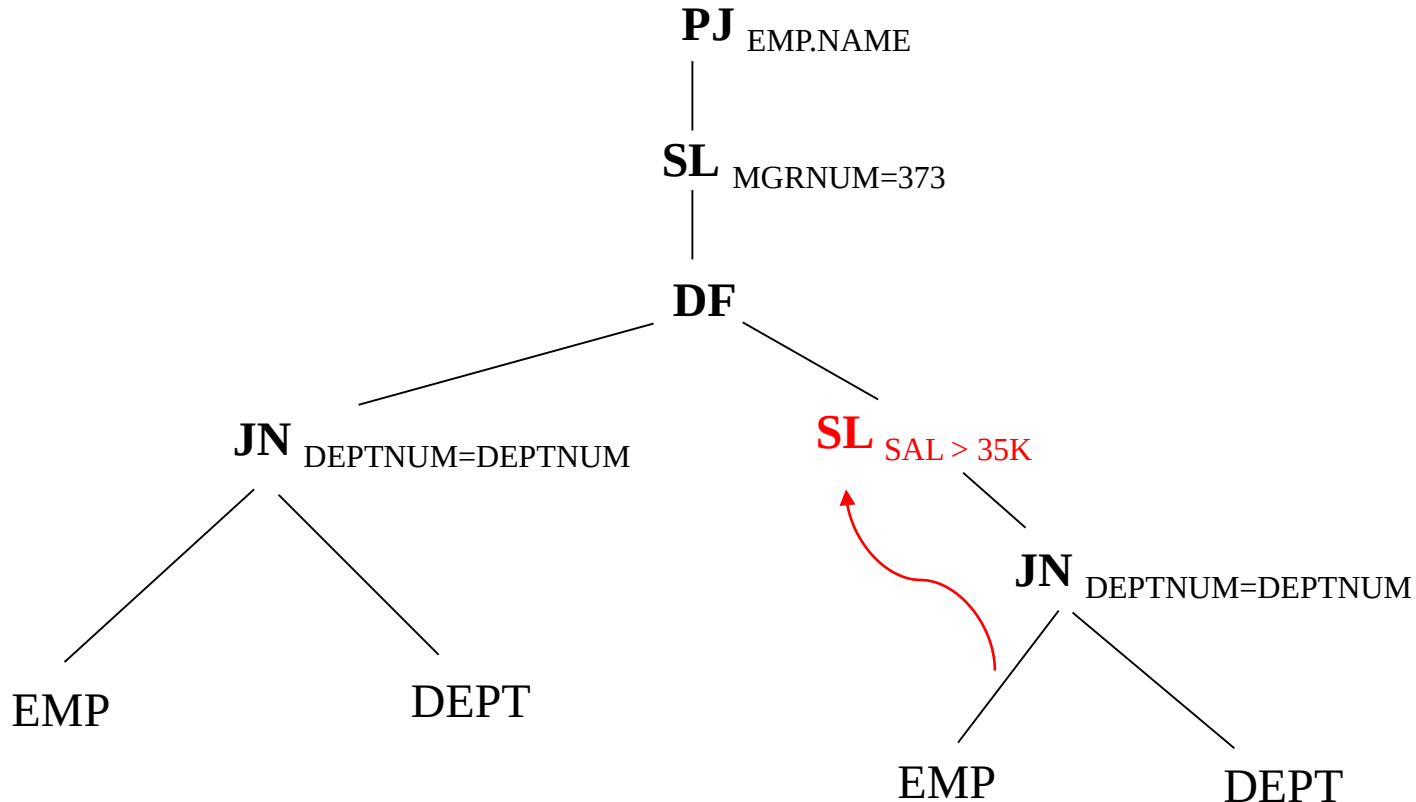
Finding Common Sub-expression

Any common portion?



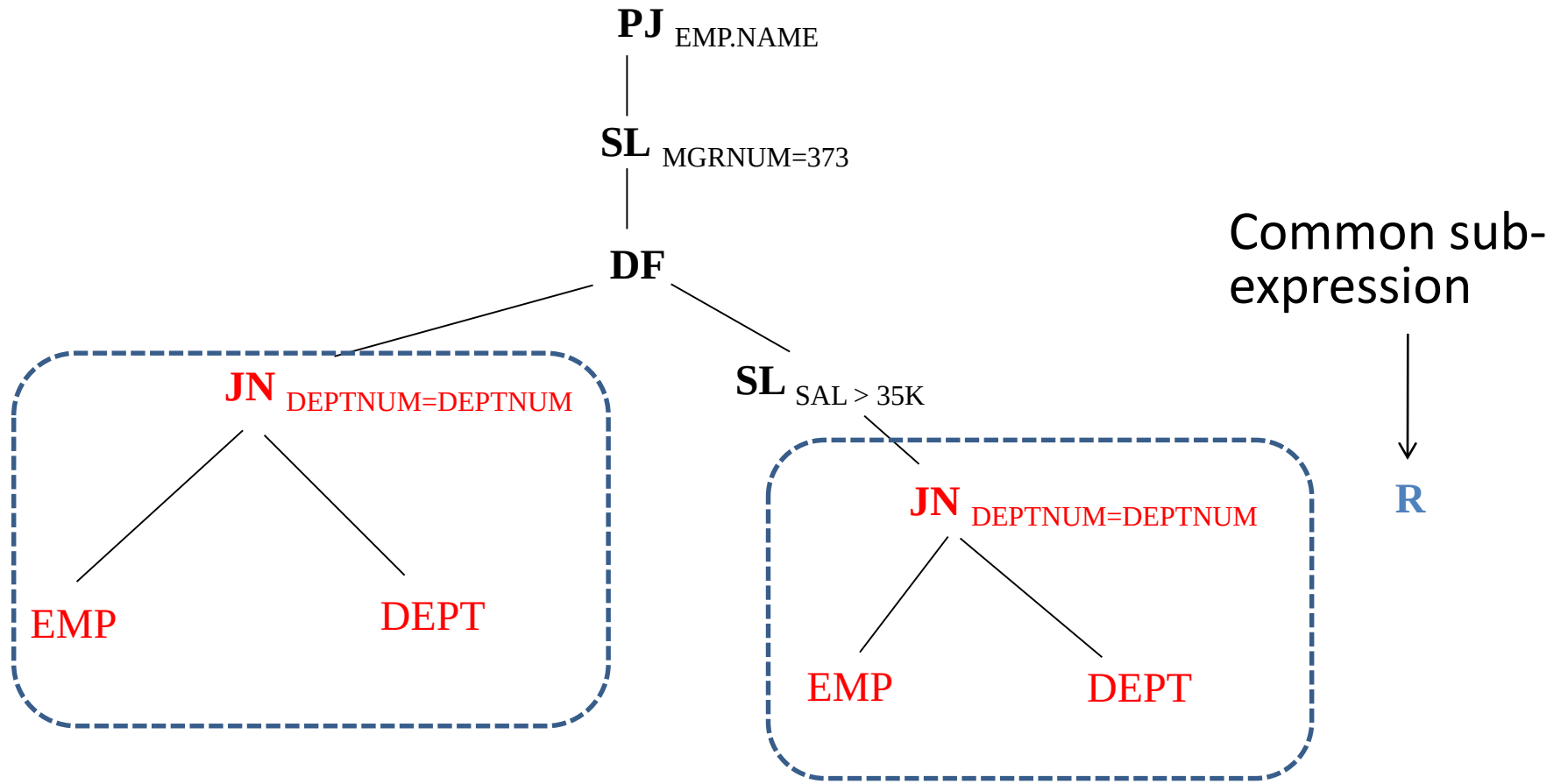
Finding Common Sub-expression

Any common portion? NOW?

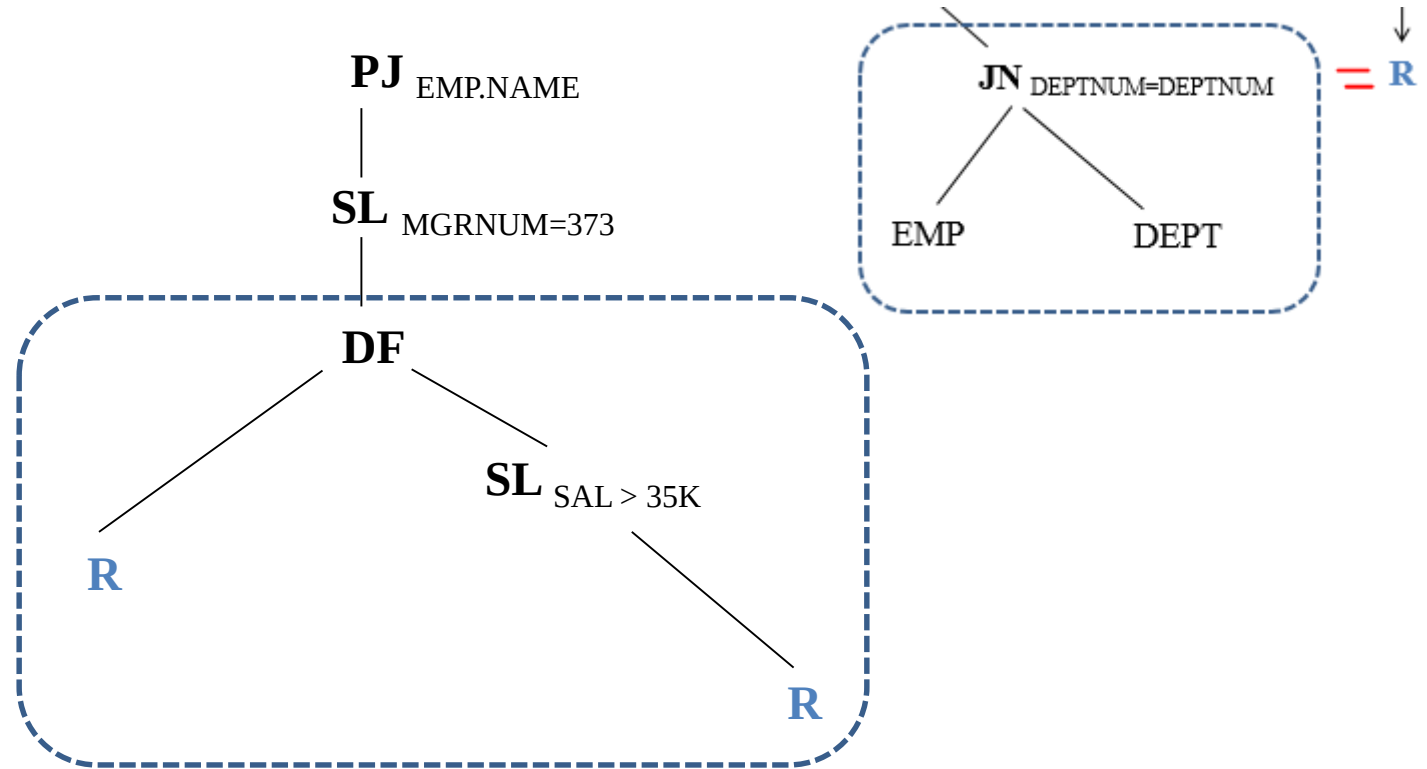


Finding Common Sub-expression

Any common portion? NOW?

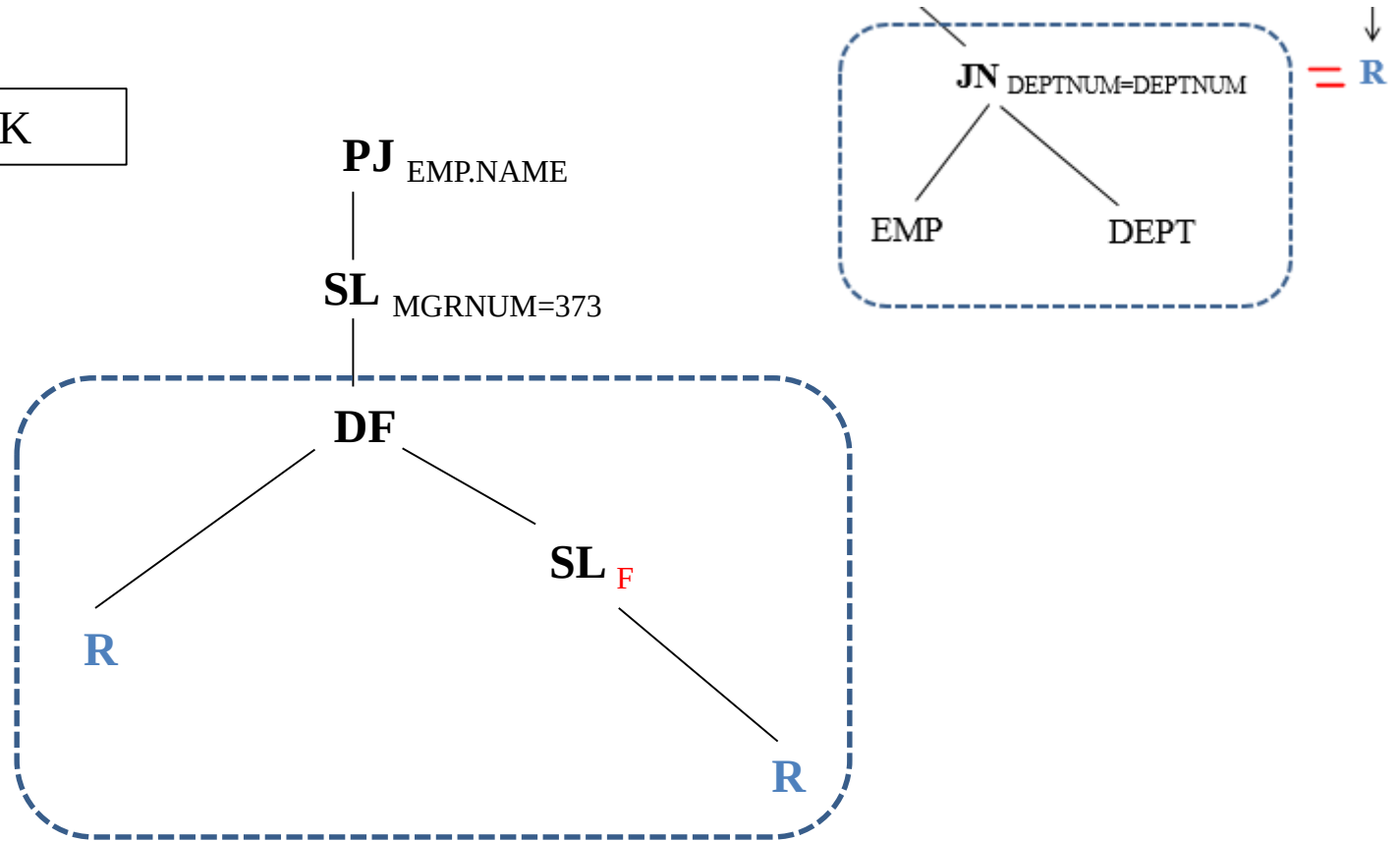


Finding Common Sub-expression



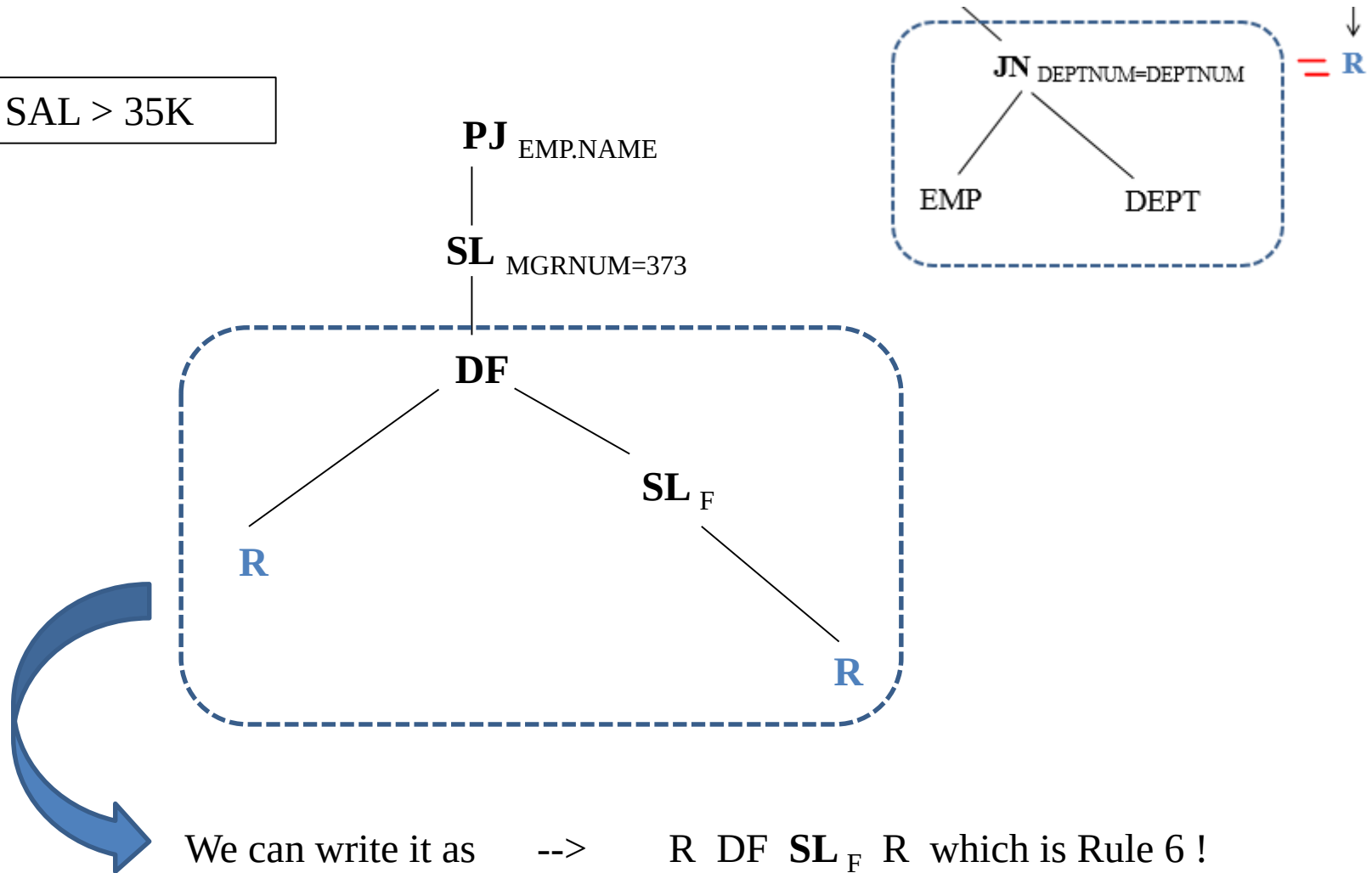
Finding Common Sub-expression

$F = \text{SAL} > 35K$



Finding Common Sub-expression

$F = \text{SAL} > 35K$

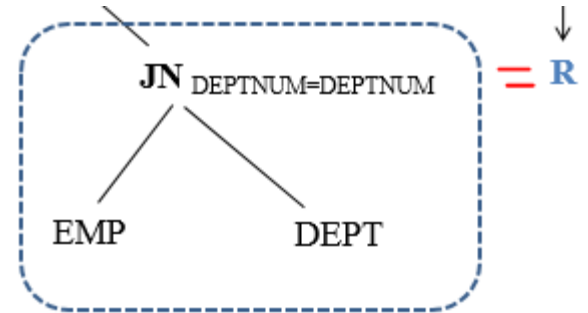


Removing Common Sub-expression

$F = \text{SAL} > 35\text{K}$

PJ EMP.NAME
|
SL MGRNUM=373

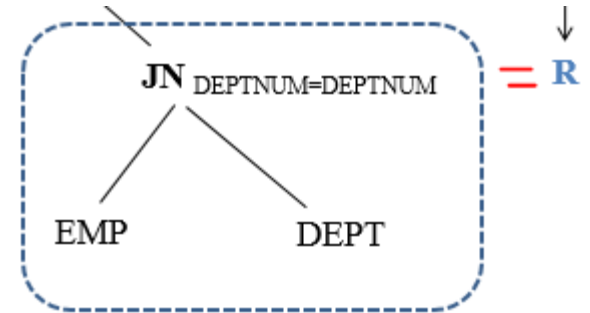
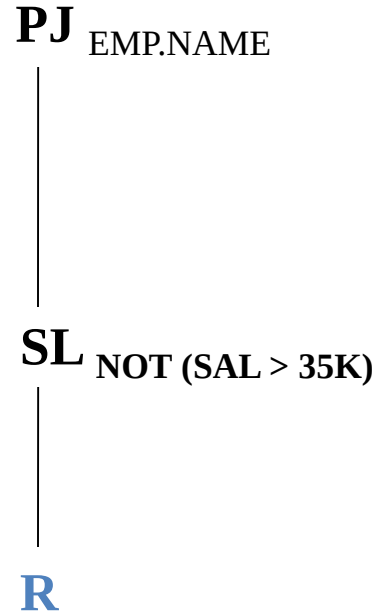
|
SL NOT F
|
R



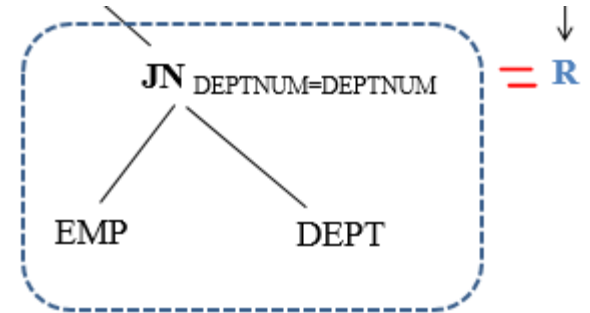
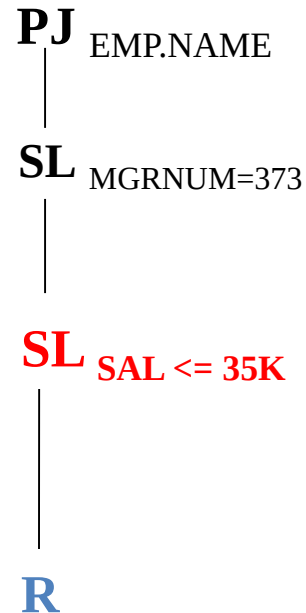
• $R \text{ DF } \text{SL}_F R \leftrightarrow \text{SL}_{\text{NOT } F} R$

Removing Common Sub-expression

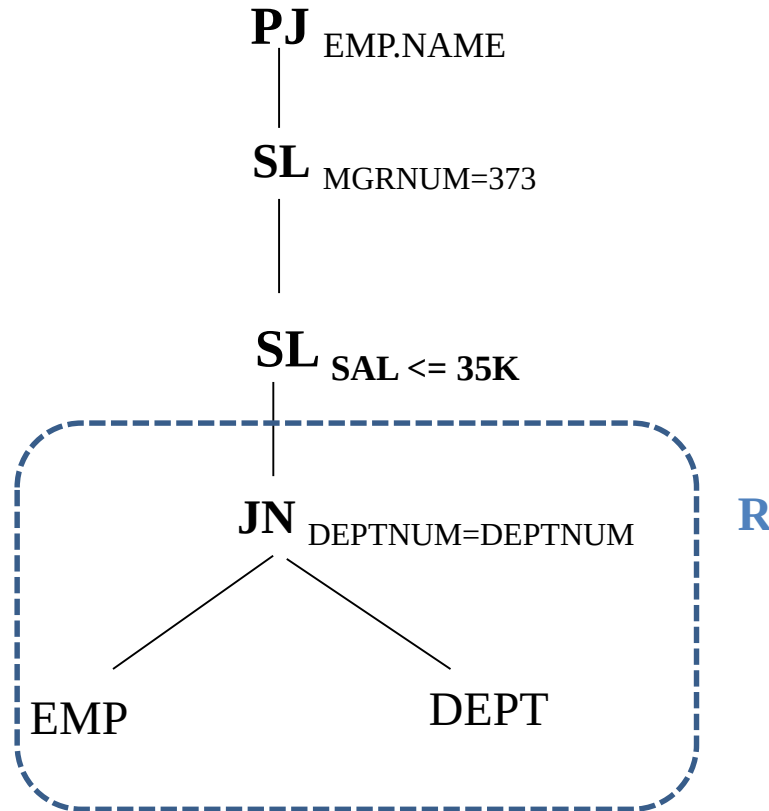
$F = \text{SAL} > 35\text{K}$



Removing Common Sub-expression



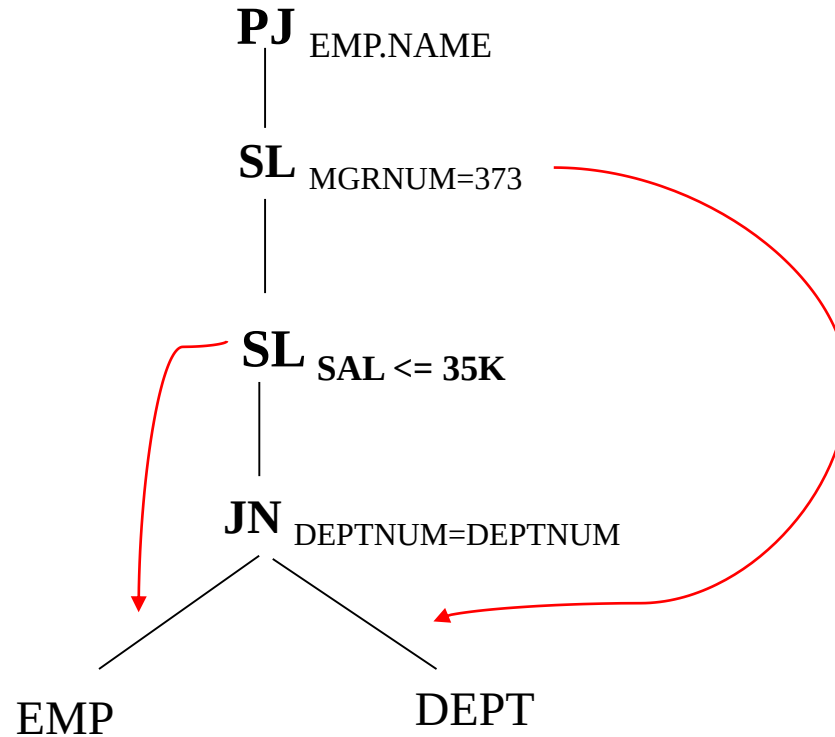
Removing Common Sub-expression



Can you apply Criterion 1 and/or 2 on this tree?

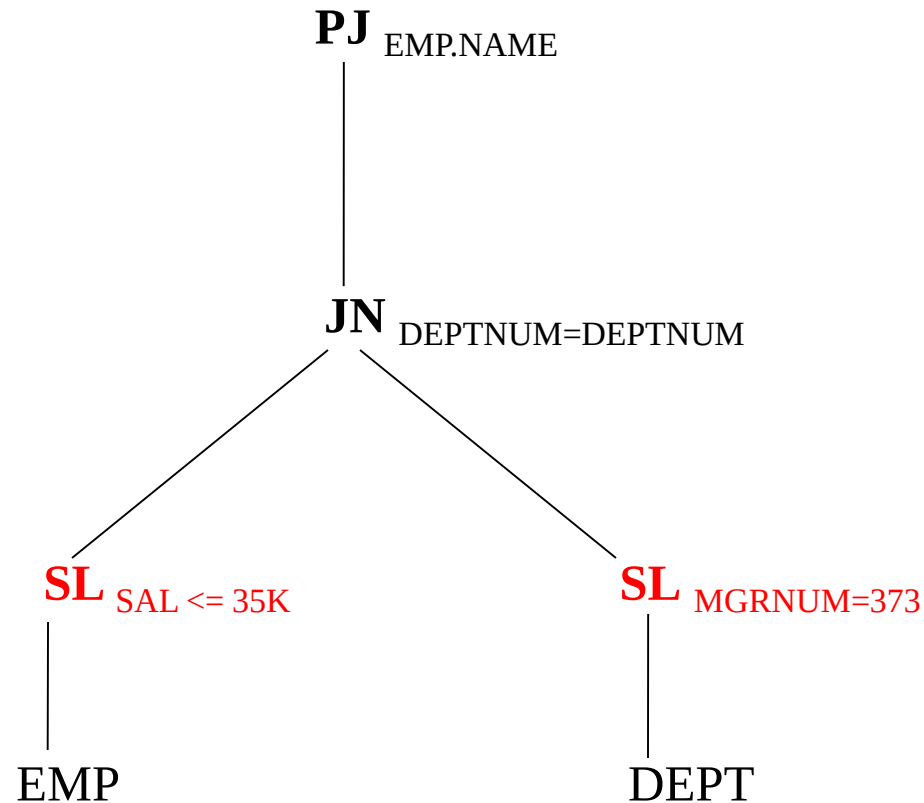
Simplification

Applying **Criterion 2** -



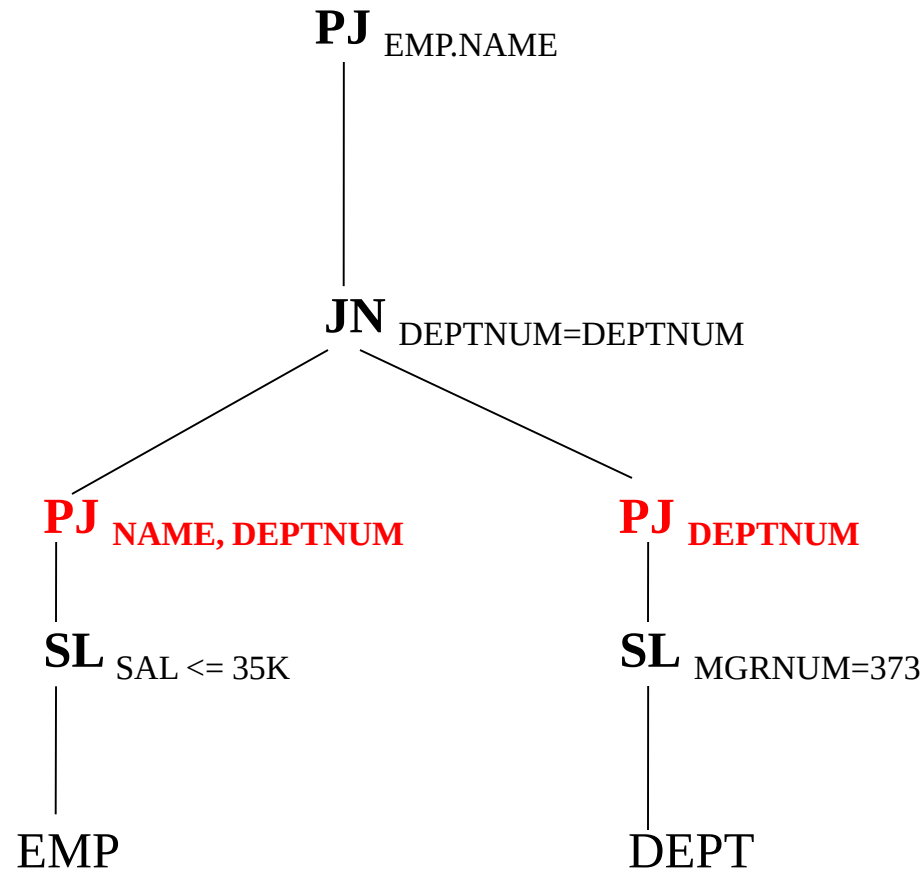
Simplification

After Applying criterion – 2



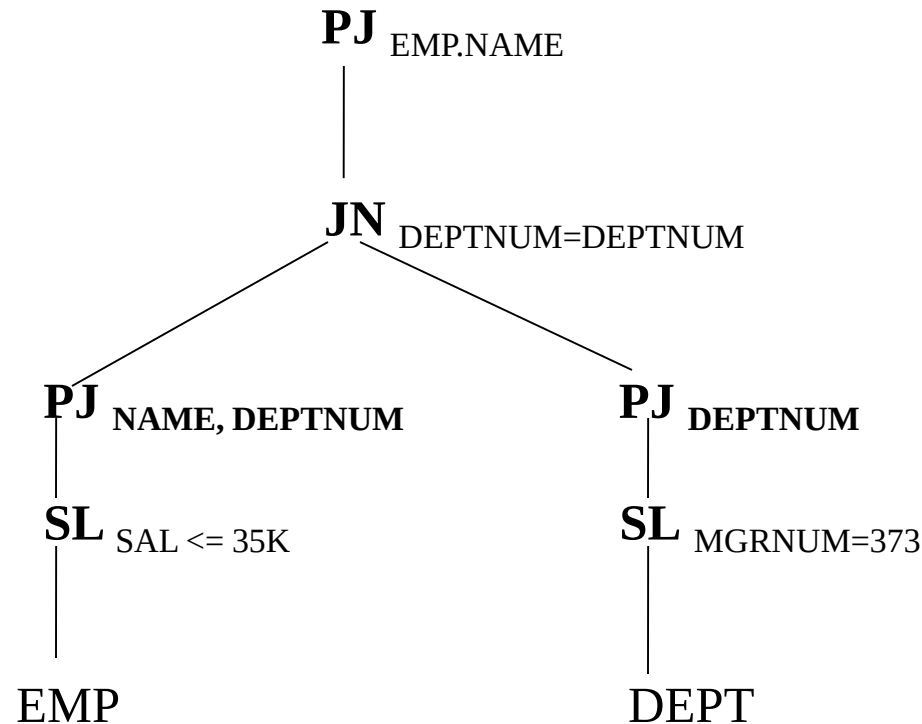
Simplification

After Applying criterion – 1



Transformed Query

Q_T: PJ_{EMP.NAME} ((PJ_{NAME,DEPTNUM} SL_{SAL<=35K} EMP) JN_{DEPTNUM=DEPTNUM} (PJ_{DEPTNUM} SL_{MGRNUM=373} DEPT))



Transformed Query

Output:

Q_T: **PJ**_{EMP.NAME} ((**PJ**_{NAME,DEPTNUM} **SL**_{SAL<=35K} *EMP*) **JN**_{DEPTNUM=DEPTNUM}
(**PJ**_{DEPTNUM} **SL**_{MGRNUM=373} *DEPT*))

Input:

Q: **PJ**_{EMP.NAME} **SL**_{MGRNUM=373} ((*EMP* **JN**_{DEPTNUM=DEPTNUM} *DEPT*)
DF (**SL**_{SAL > 35K} *EMP* **JN**_{DEPTNUM=DEPTNUM} *DEPT*))

$$Q \longleftrightarrow Q_T$$

Example 2.2

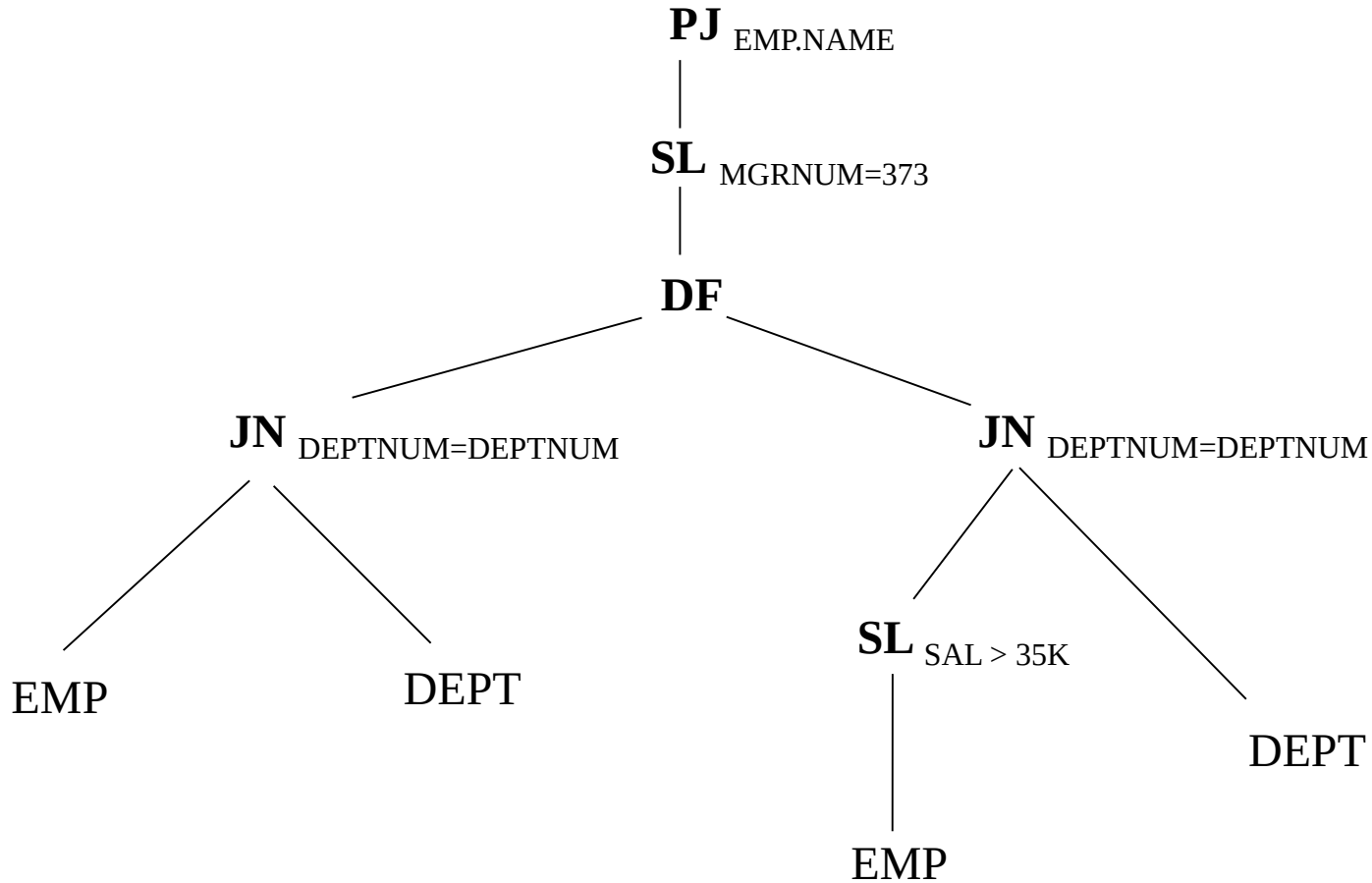
EMP (EMPNUM, DEPTNUM, NAME, SAL, AGE)
DEPT (DEPTNUM, NAME, AREA, MGRNUM)

Q: PJ *EMP.NAME* **SL** *MGRNUM=373* ((*EMP JN* *DEPTNUM=DEPTNUM* *DEPT*) **DF** (*SL* *SAL > 35K* *EMP JN* *DEPTNUM=DEPTNUM* *DEPT*))

Now, answer the following questions.

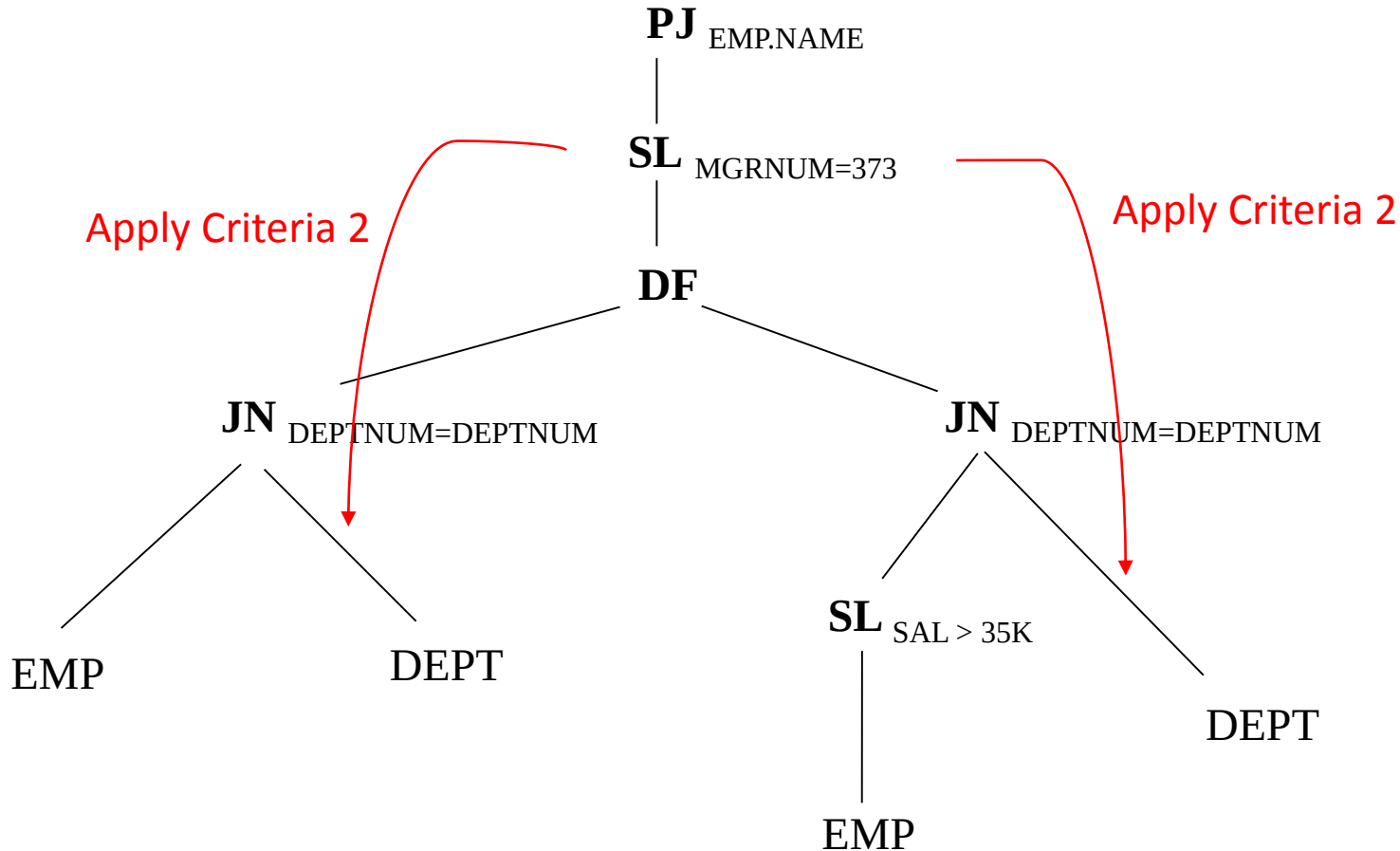
- i. Draw the *operator tree*. [2]
- ii. Perform step-by-step transformations to simplify the operator tree, indicating which rule and criterion is applied at each step. [5]

Operator Tree



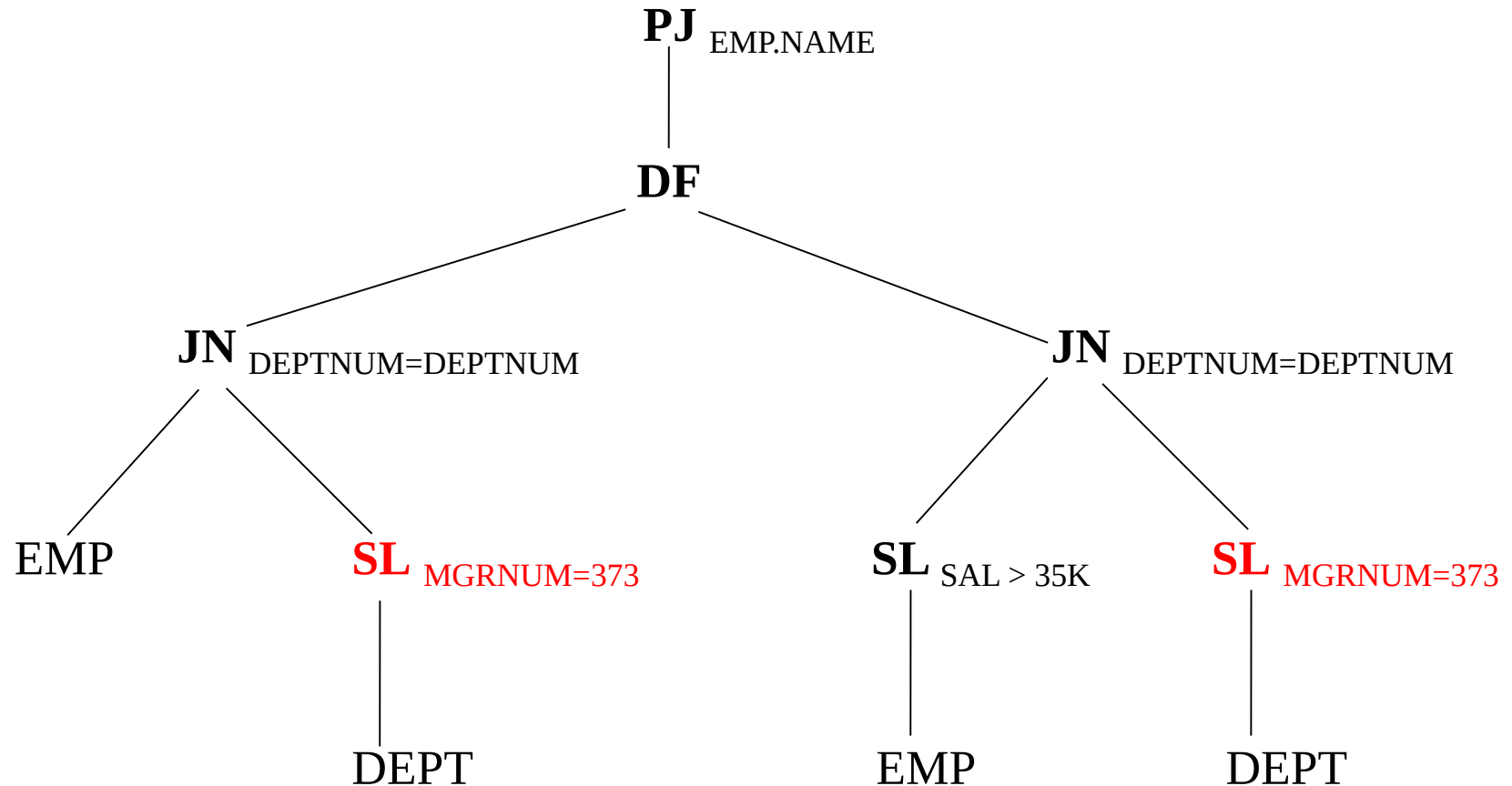
Finding Common Sub-expression

Any common portion?



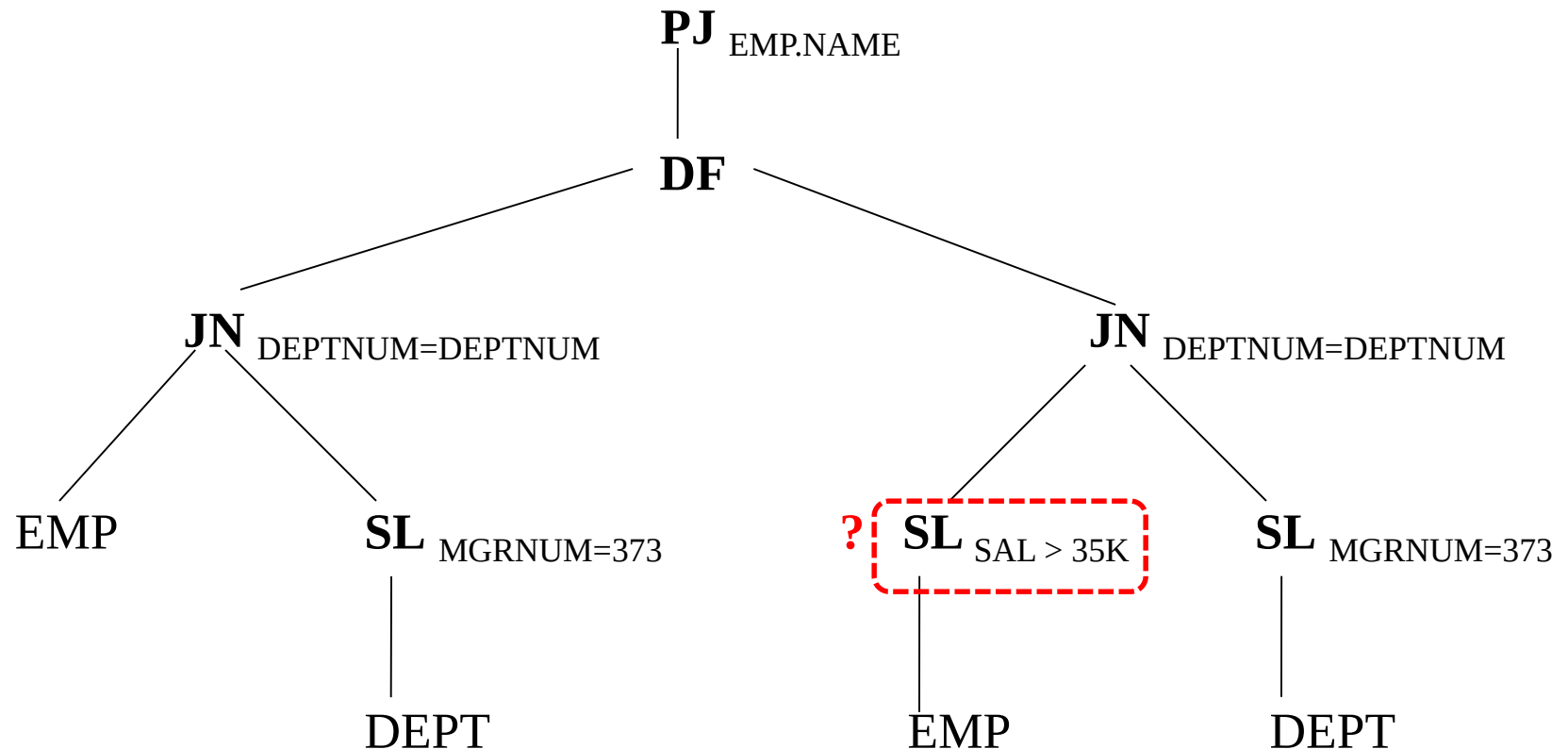
Finding Common Sub-expression

Any common portion?



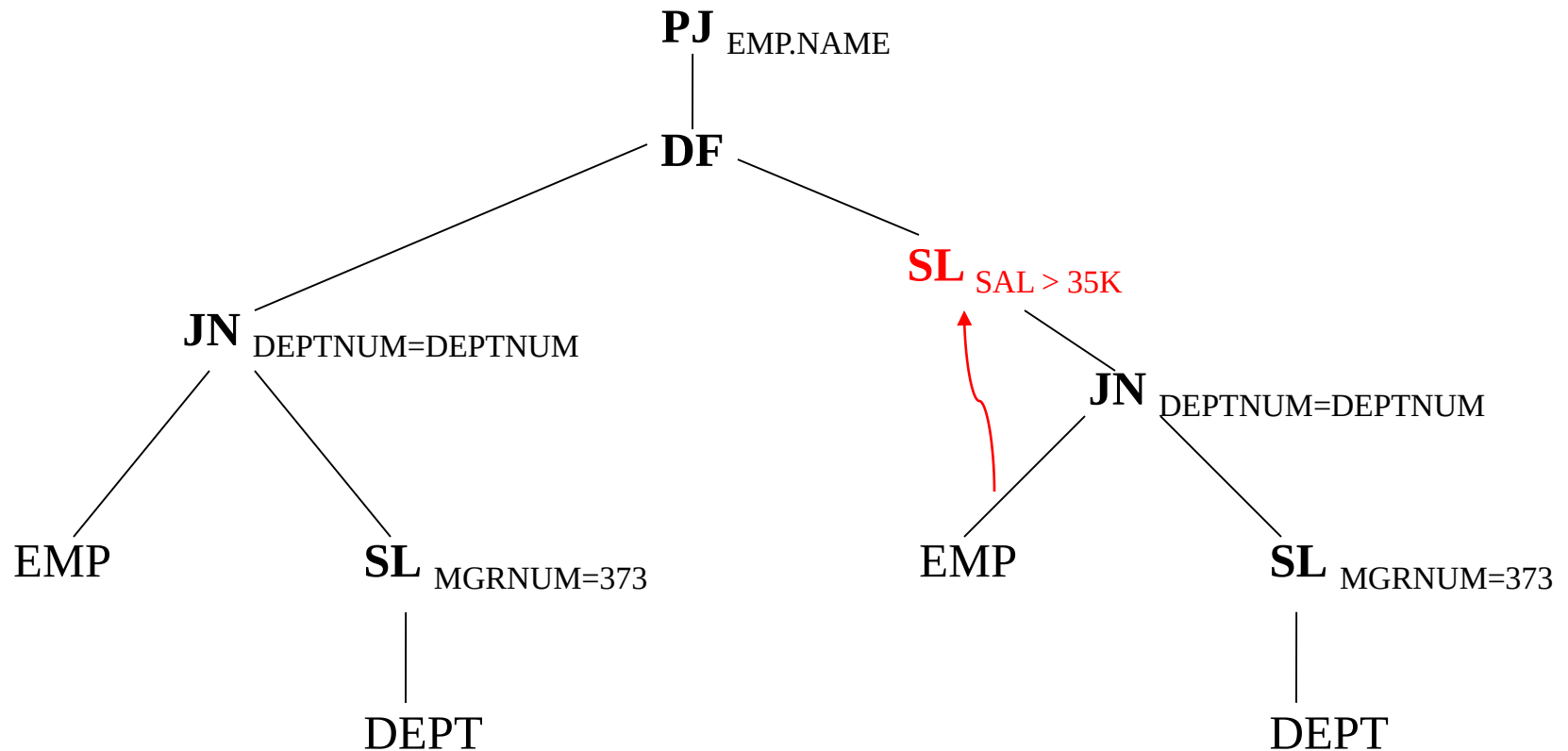
Finding Common Sub-expression

Any common portion?

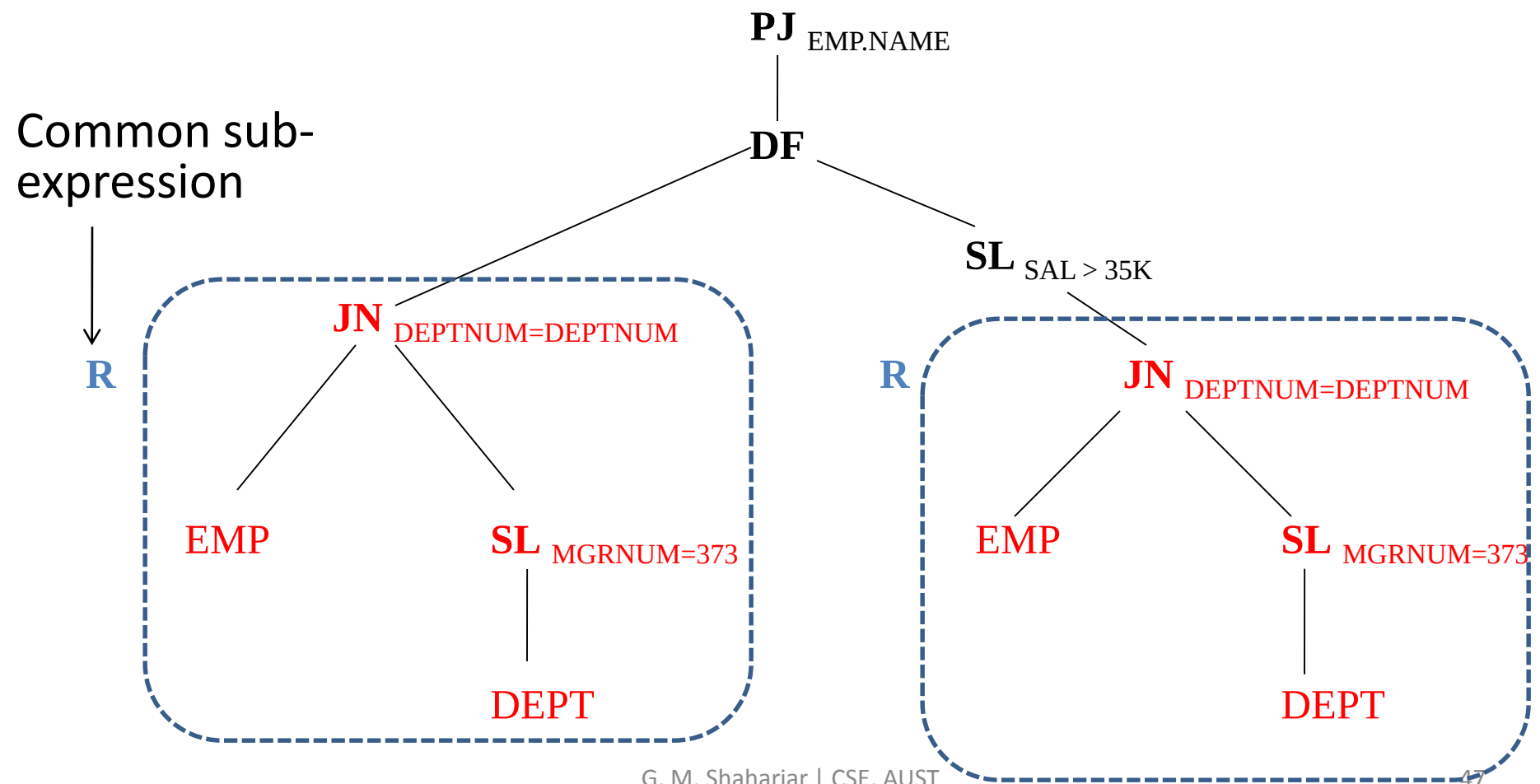


Finding Common Sub-expression

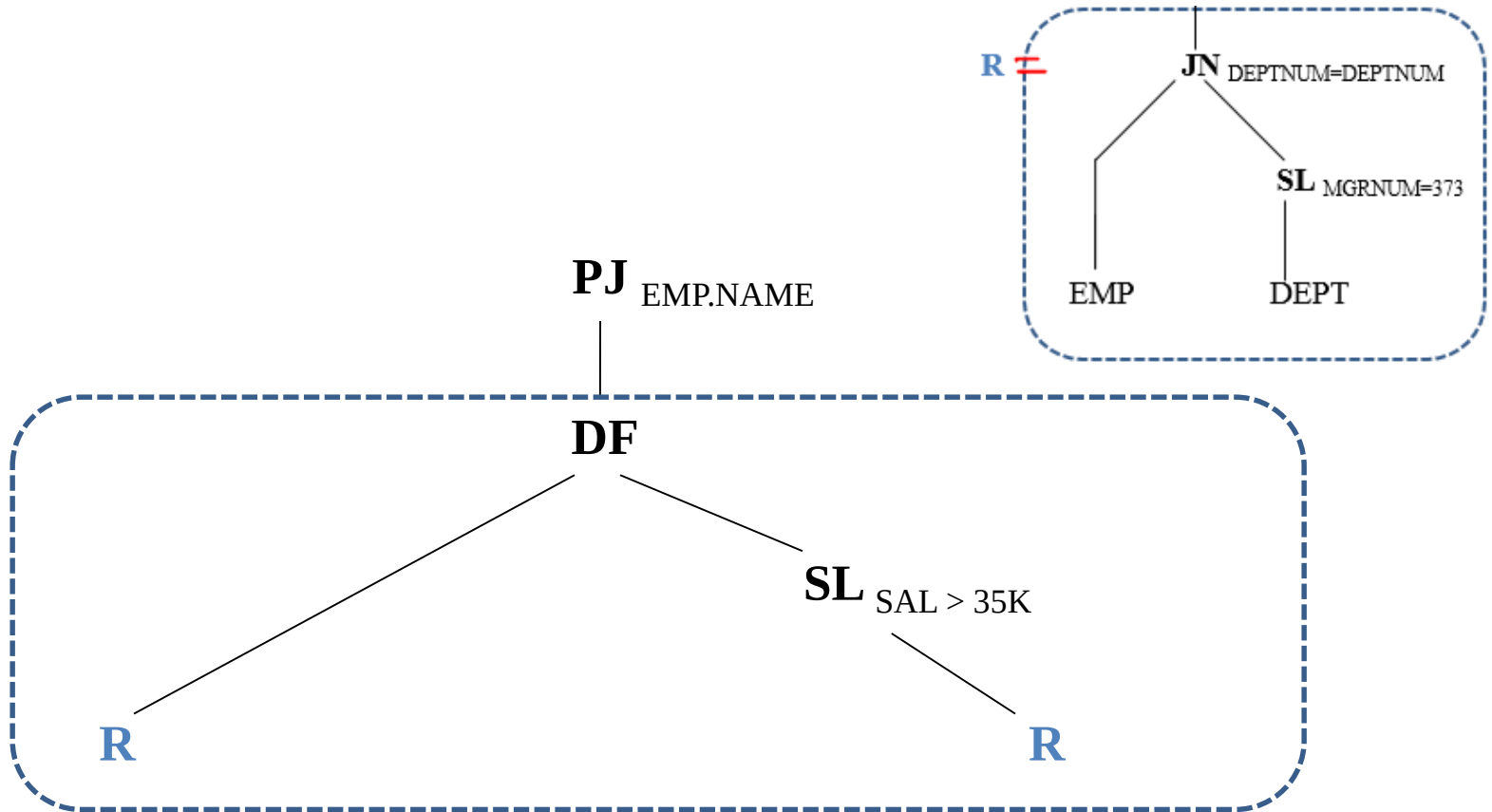
Any common portion? NOW?



Finding Common Sub-expression

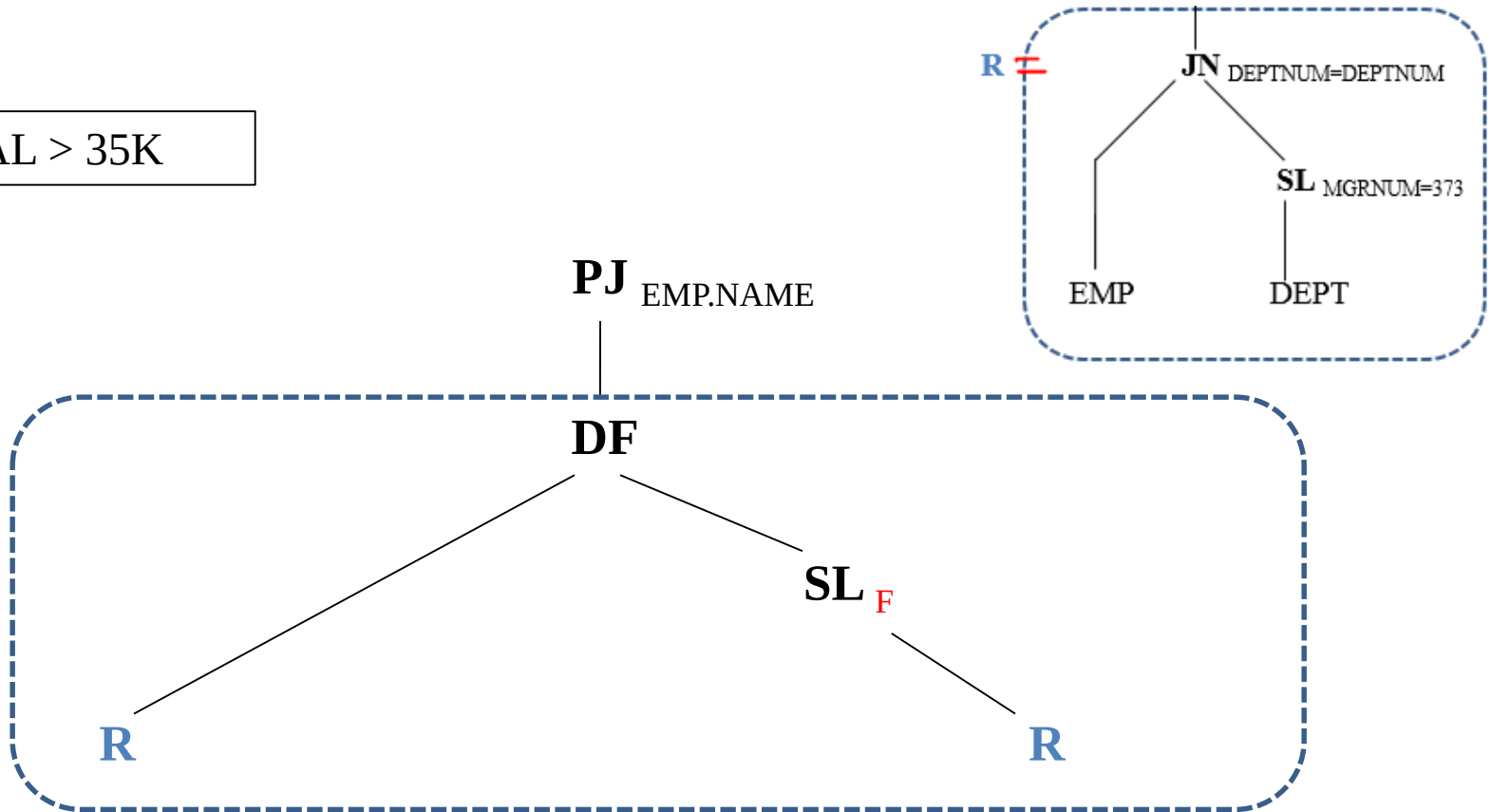


Finding Common Sub-expression



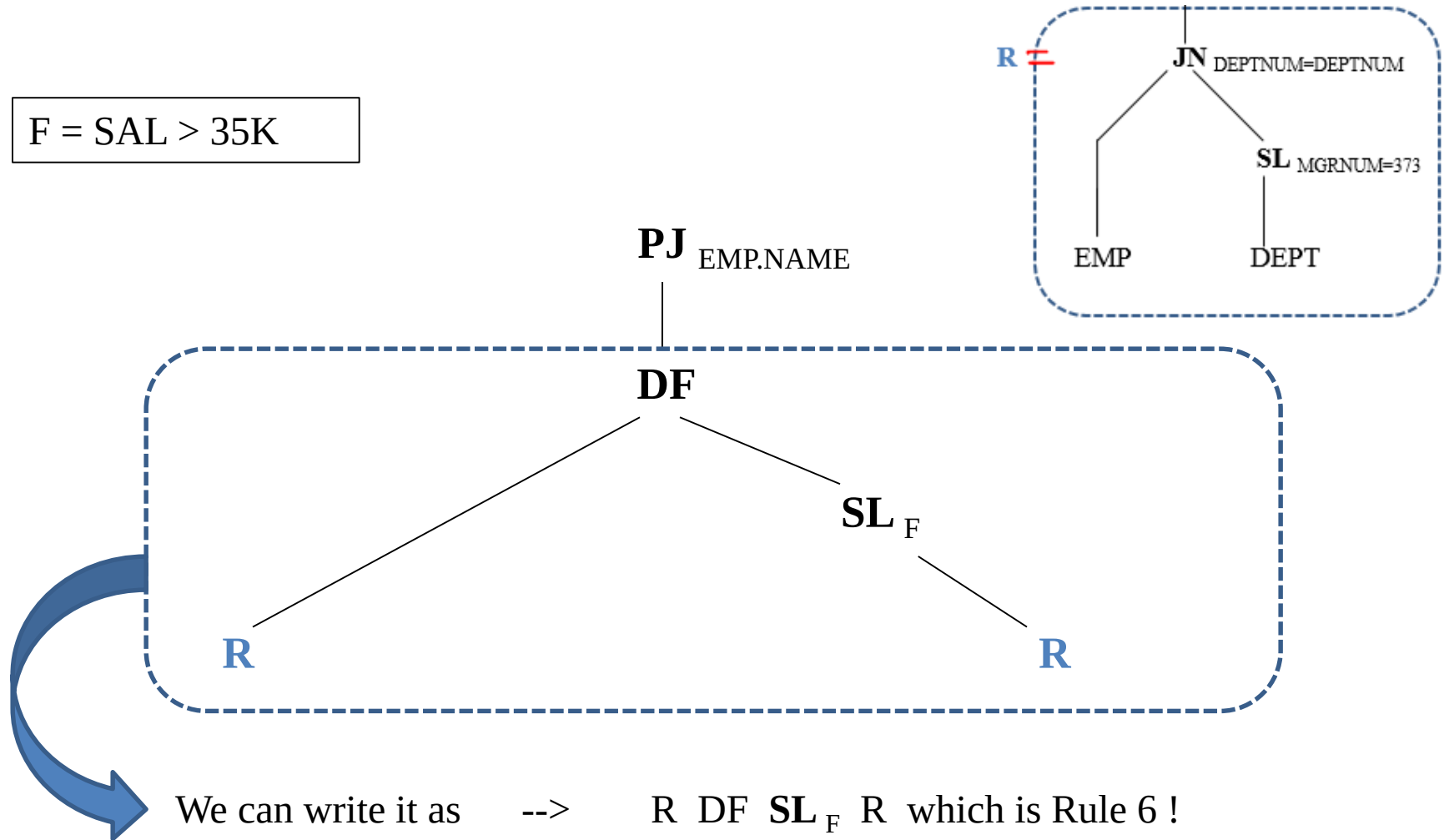
Finding Common Sub-expression

$F = \text{SAL} > 35\text{K}$



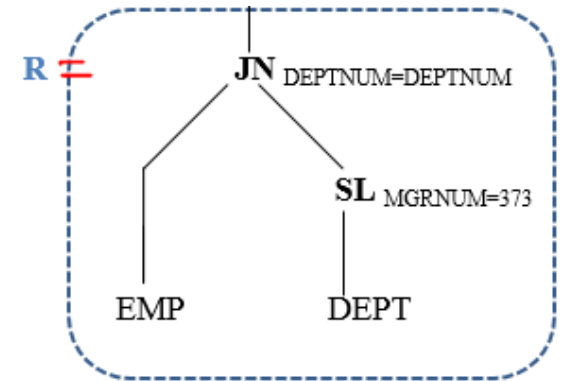
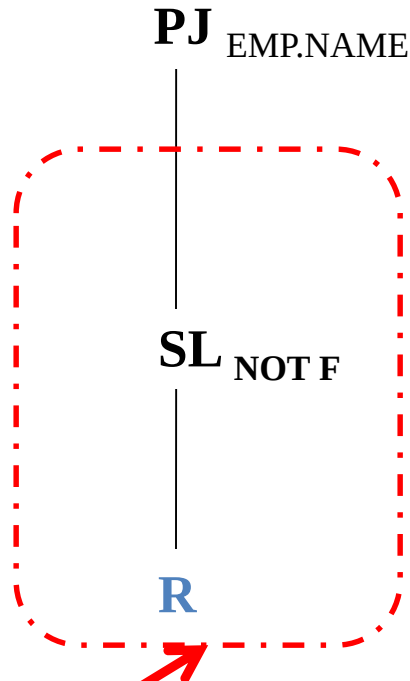
Finding Common Sub-expression

$F = \text{SAL} > 35\text{K}$



Removing Common Sub-expression

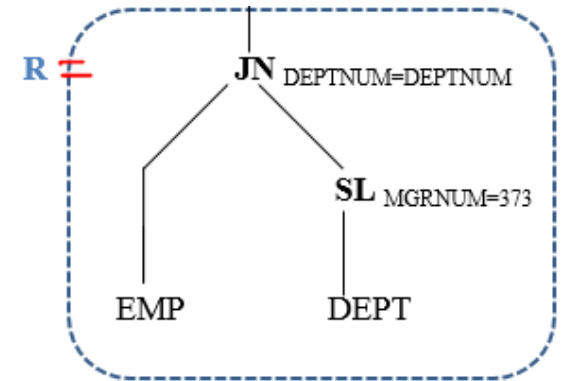
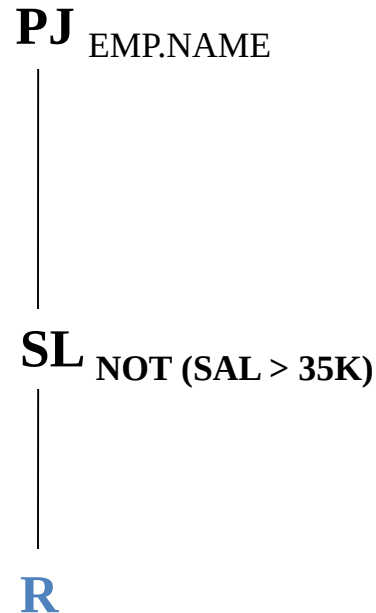
$F = \text{SAL} > 35K$



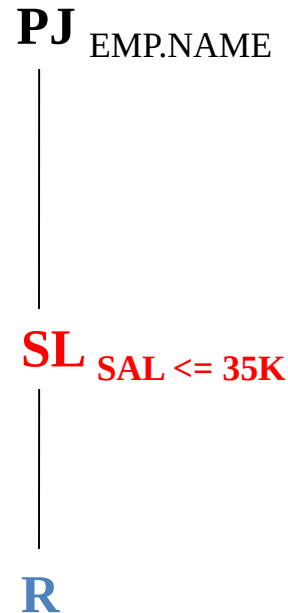
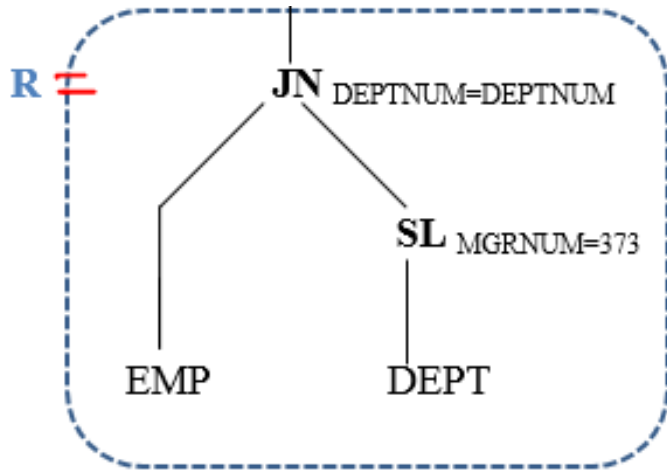
• $R \text{ DF } SL_F R \leftrightarrow SL_{NOT F} R$

Removing Common Sub-expression

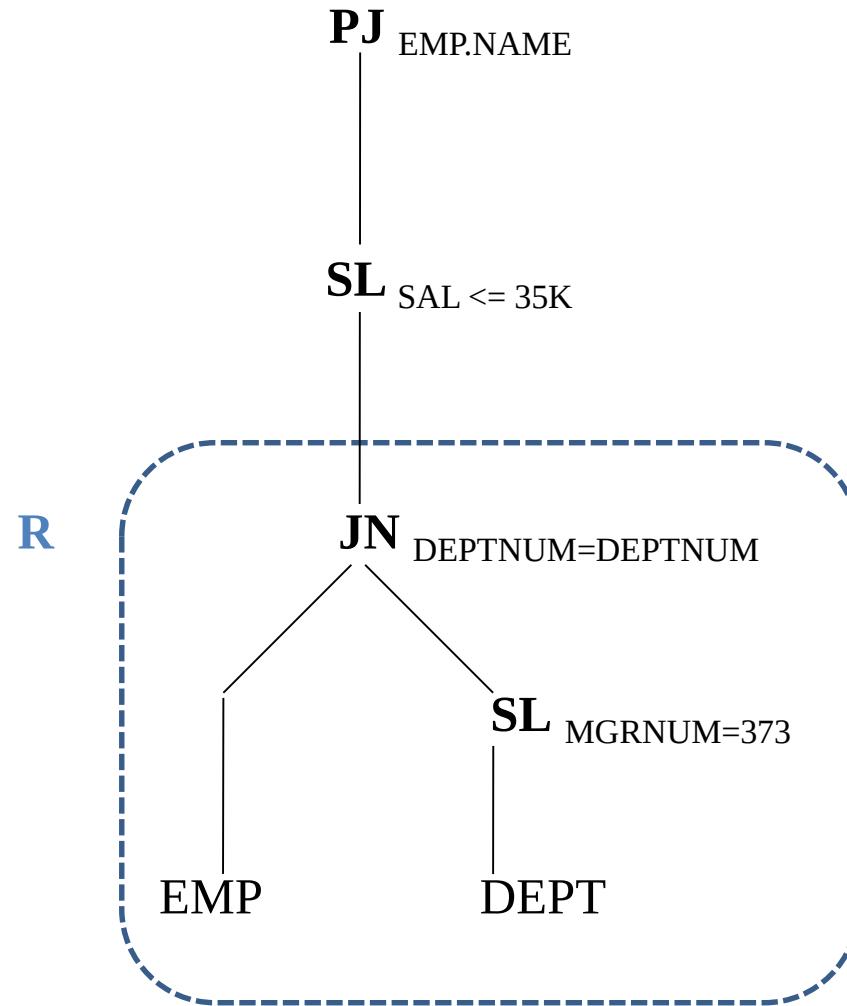
$F = \text{SAL} > 35\text{K}$



Removing Common Sub-expression



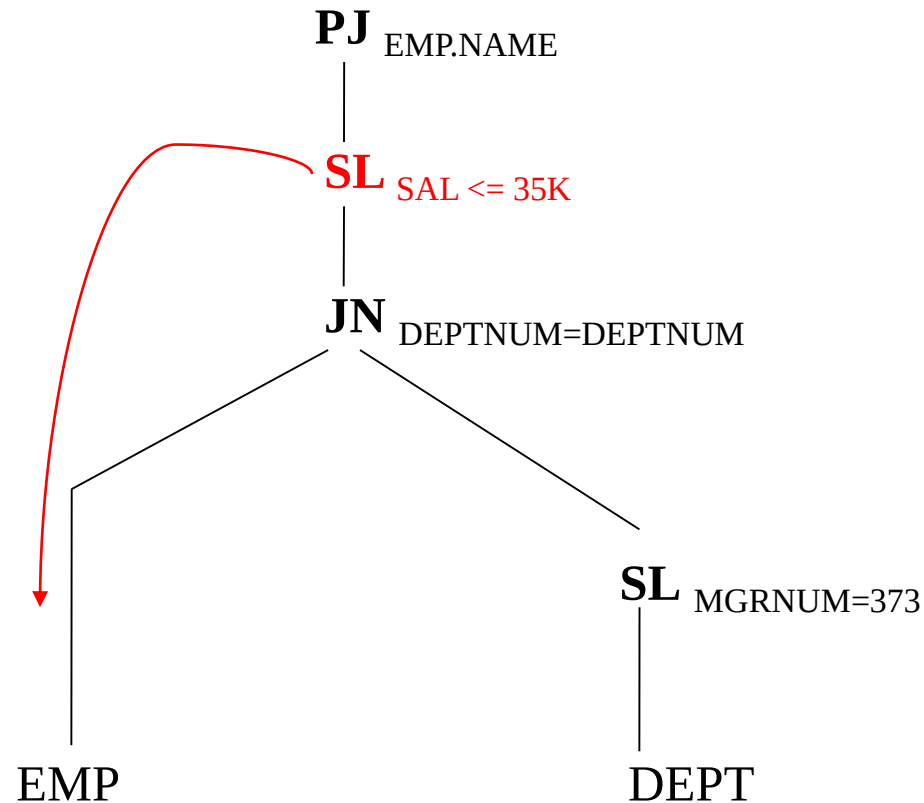
Removing Common Sub-expression



Can you apply Criterion 1 and/or 2 on this tree?

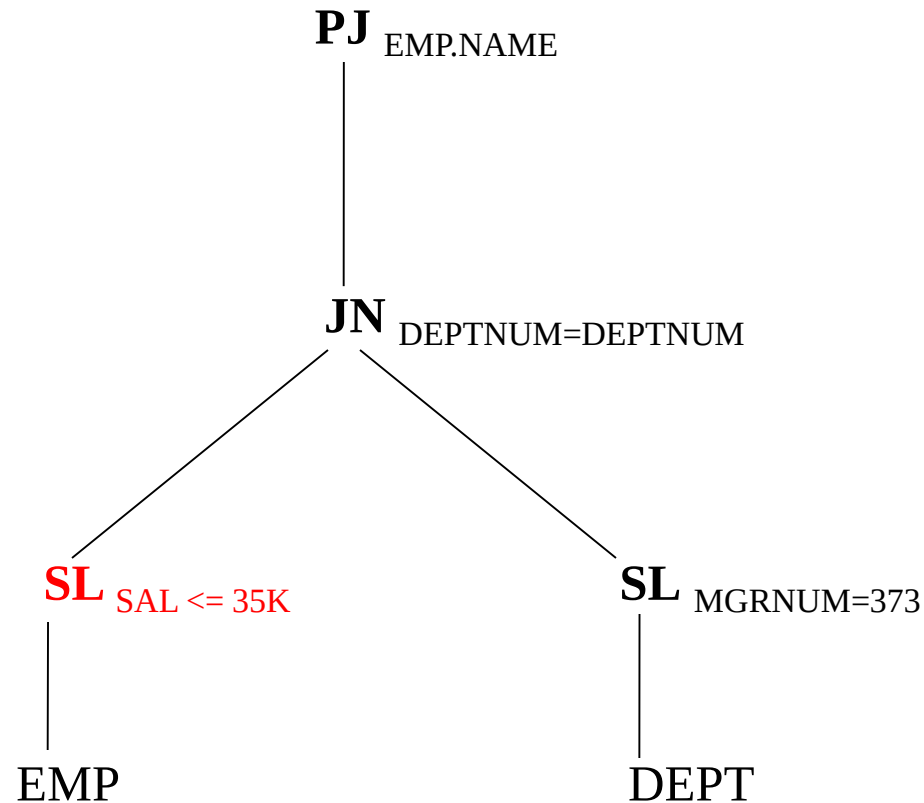
Simplification

Applying criterion – 2



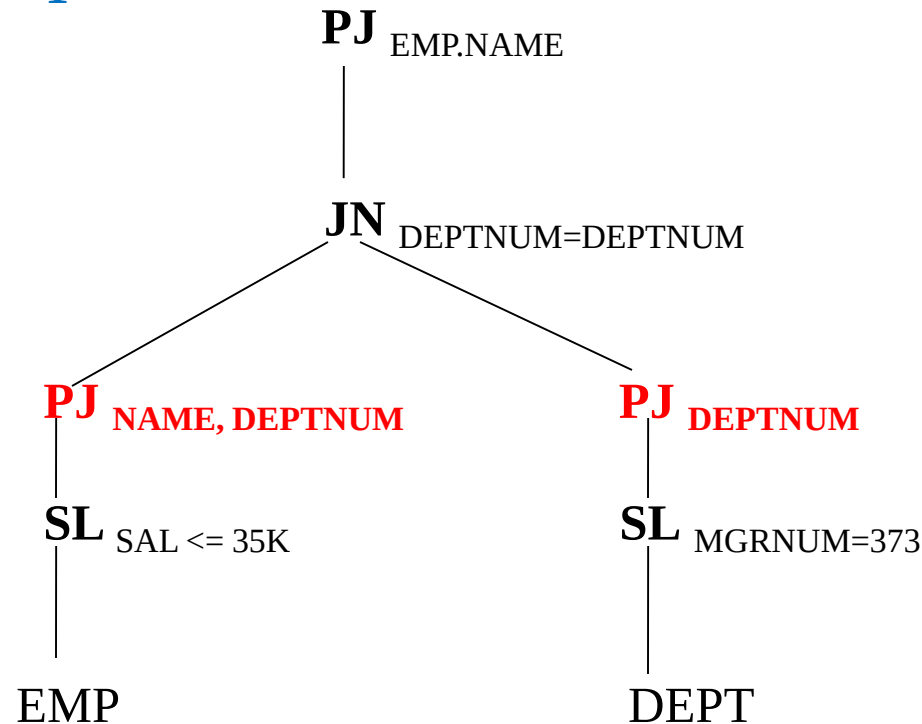
Simplification

After Applying criterion – 2



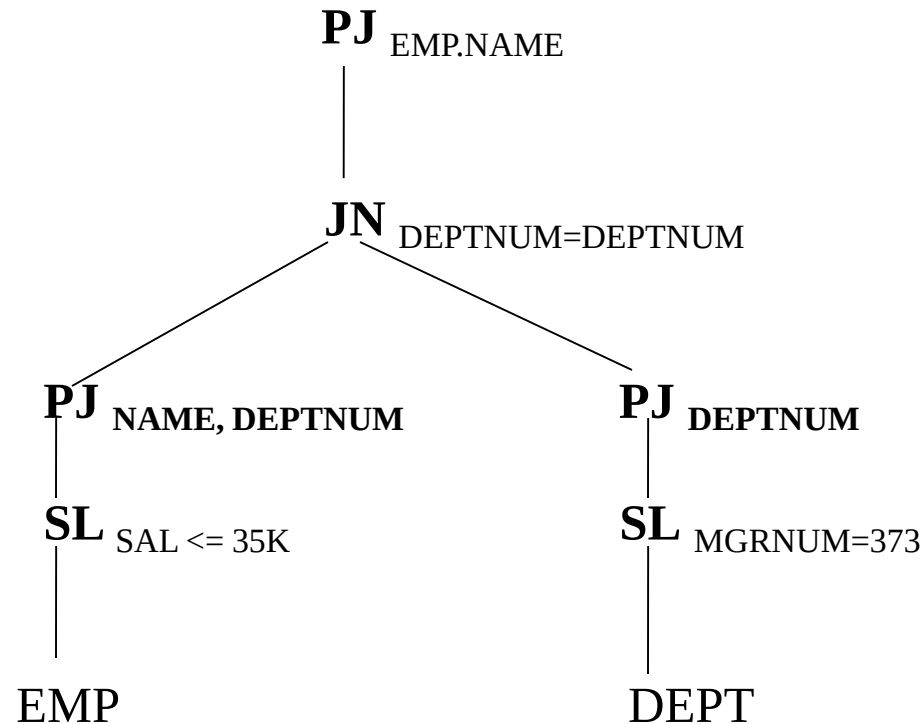
Simplification

After Applying criterion – 1



Transformed Query

Q_T: PJ_{EMP.NAME} ((PJ_{NAME,DEPTNUM} SL_{SAL<=35K} EMP) JN_{DEPTNUM=DEPTNUM} (PJ_{DEPTNUM} SL_{MGRNUM=373} DEPT))



Transformed Query

Output:

Q_T: **PJ**_{EMP.NAME} ((**PJ**_{NAME,DEPTNUM} **SL**_{SAL<=35K} *EMP*) **JN**_{DEPTNUM=DEPTNUM}
(**PJ**_{DEPTNUM} **SL**_{MGRNUM=373} *DEPT*))

Input:

Q: **PJ**_{EMP.NAME} **SL**_{MGRNUM=373} ((*EMP* **JN**_{DEPTNUM=DEPTNUM} *DEPT*)
DF (**SL**_{SAL > 35K} *EMP* **JN**_{DEPTNUM=DEPTNUM} *DEPT*))

$$Q \longleftrightarrow Q_T$$

Example 3

Practise

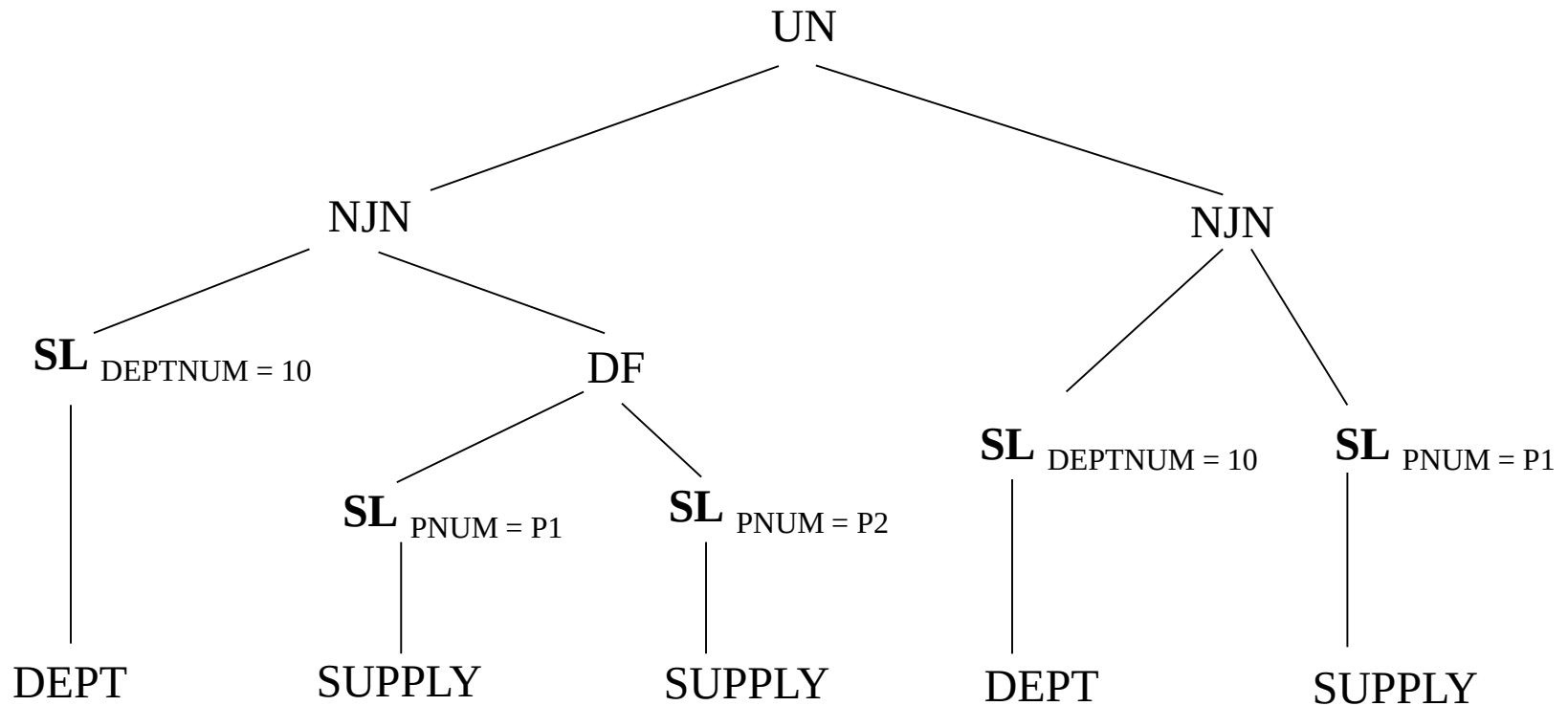
* Draw Operator Tree for the following queries:

①
SUPPLY (SNUM, PNUM, DEPTNUM, QUAN)
DEPT (DEPTNUM, NAME, AREA, MGRNUM)

Query:

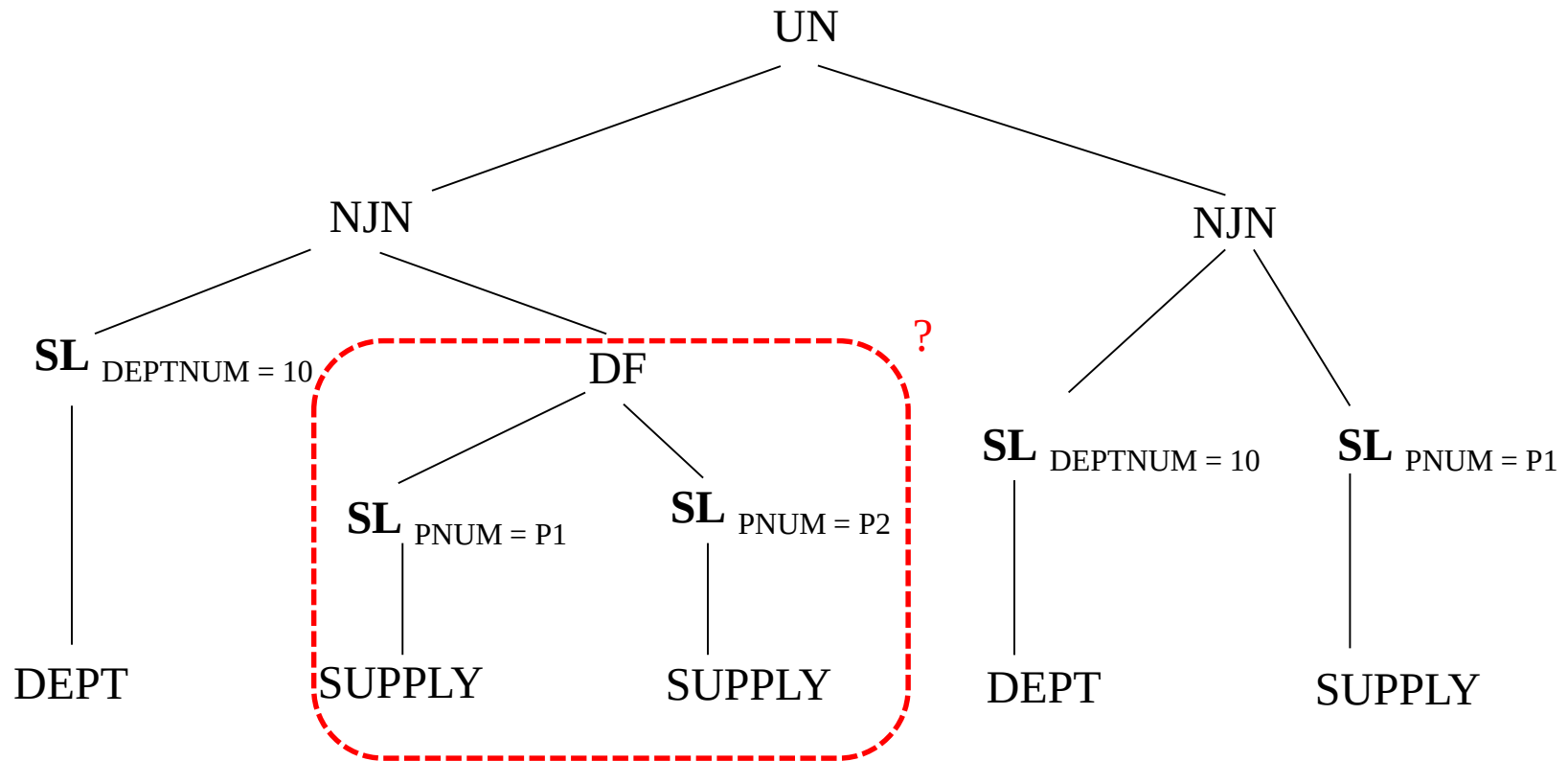
$(\sigma_{DEPTNUM=10} DEPT \bowtie (\sigma_{PNUM="P_1"} SUPPLY$
 $\bowtie \sigma_{PNUM="P_2"} SUPPLY)) \cup (\sigma_{DEPTNUM=10} DEPT$
 $\bowtie \sigma_{PNUM="P_1"} SUPPLY)$

Operator Tree



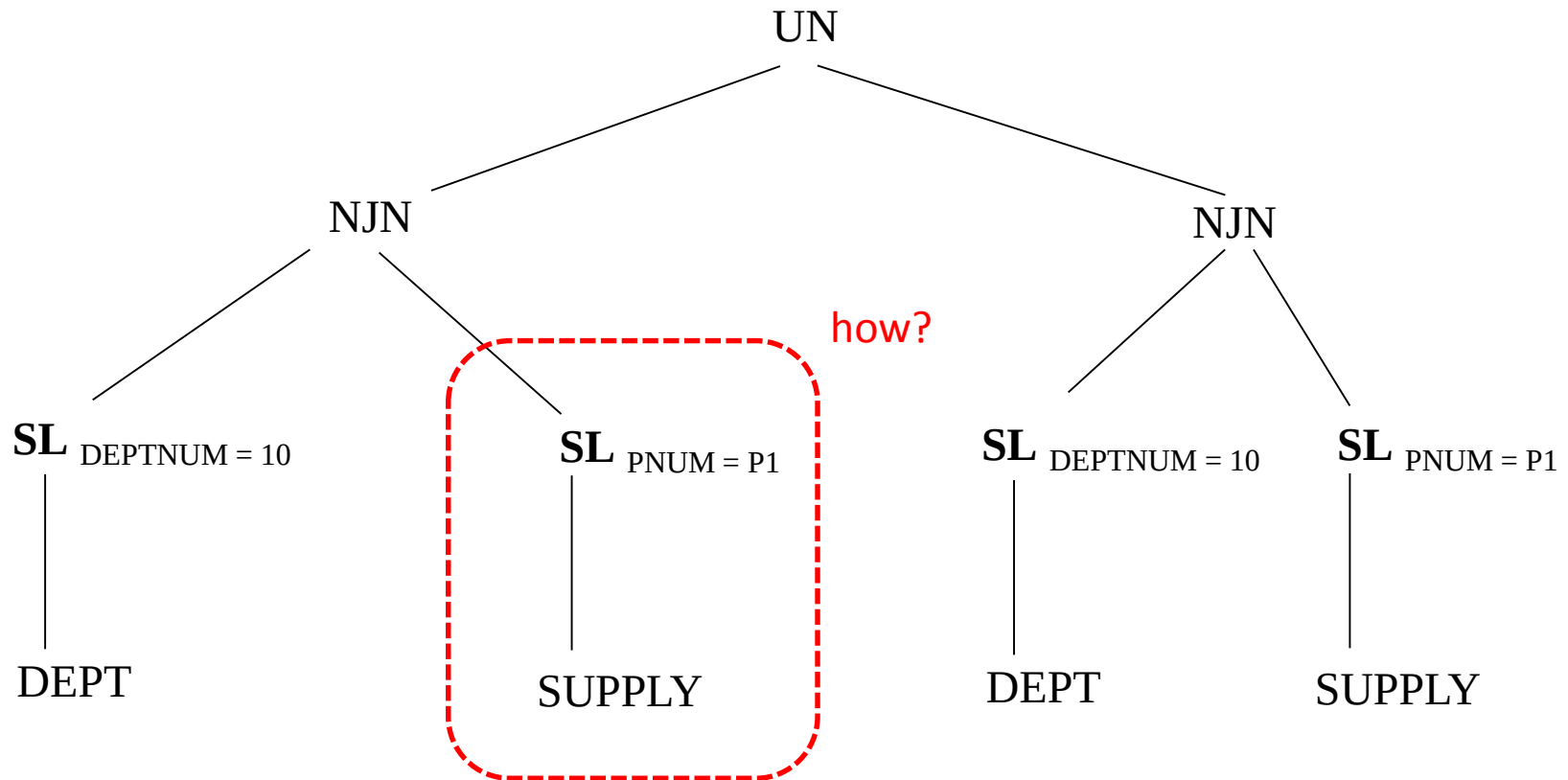
Finding Common Sub - Expression

Any common portion?



Finding Common Sub - Expression

Any common portion?



How?

SUPPLY

SNUM	PNUM	DEPTNUM	QUAN
1	P1	1	10
2	P2	2	20
3	P1	1	30
4	P2	1	40
5	P1	2	50
6	P2	1	60

A

SL_{PNUM = P1} SUPPLY

SNUM	PNUM	DEPTNUM	QUAN
1	P1	1	10
3	P1	1	30
5	P1	2	50

B

SL_{PNUM = P2} SUPPLY

SNUM	PNUM	DEPTNUM	QUAN
2	P2	2	20
4	P2	1	40
6	P2	1	60

A

DF

B

=

A

SNUM	PNUM	DEPT NUM	QUAN
1	P1	1	10
3	P1	1	30
5	P1	2	50

-

SNUM	PNUM	DEPT NUM	QUAN
2	P2	2	20
4	P2	1	40
6	P2	1	60

=

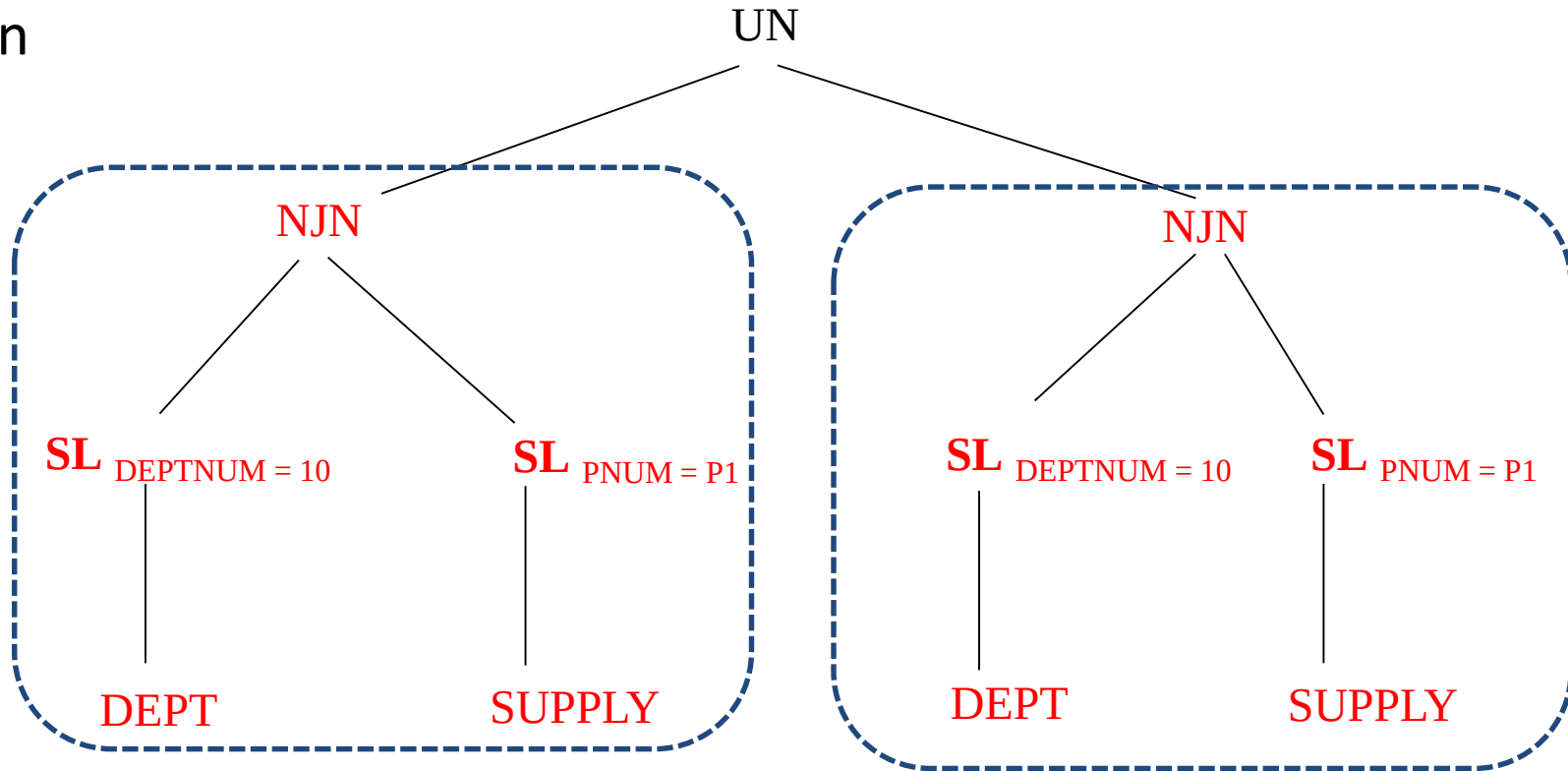
SNUM	PNUM	DEPT NUM	QUAN
1	P1	1	10
3	P1	1	30
5	P1	2	50

Finding Common Sub - Expression

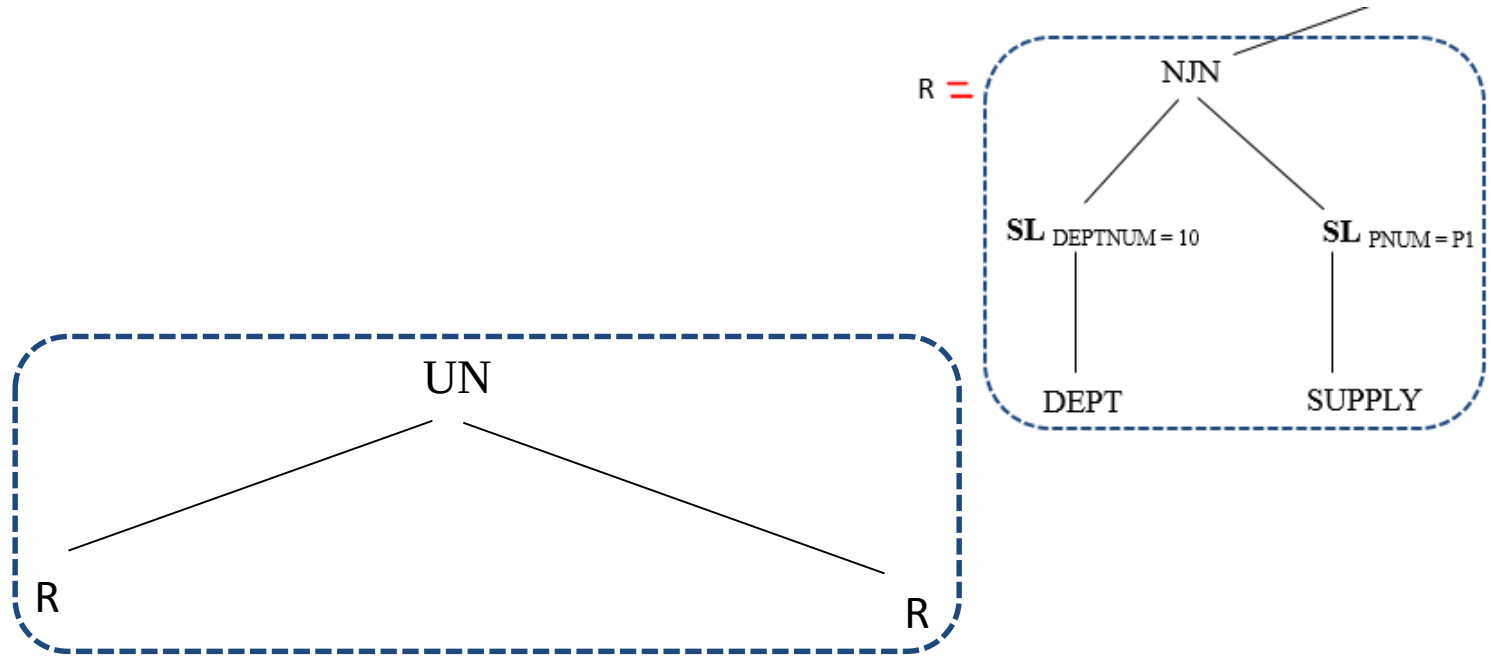
Any common portion? NOW?

Common sub-expression

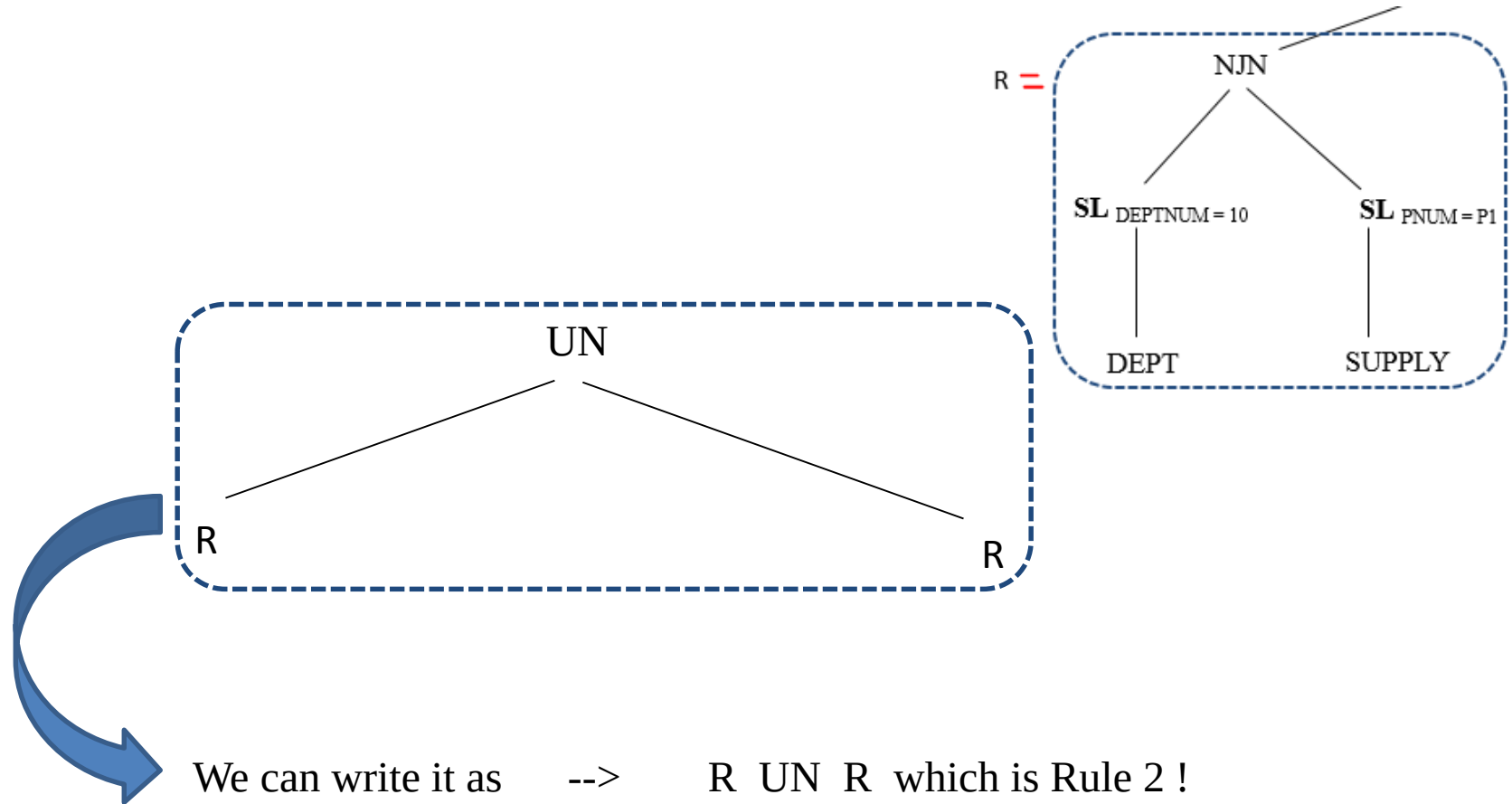
R



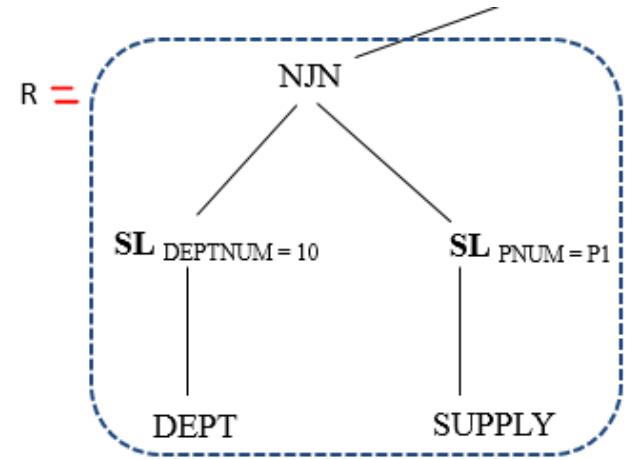
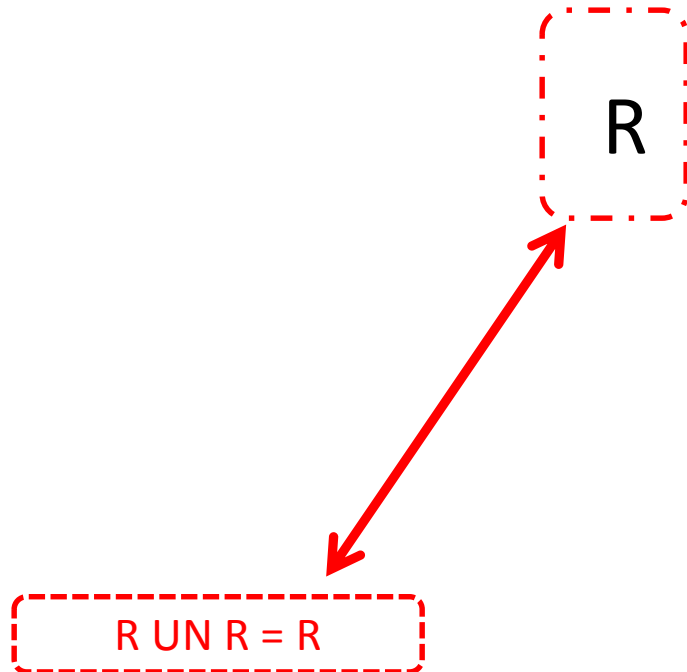
Finding Common Sub - Expression



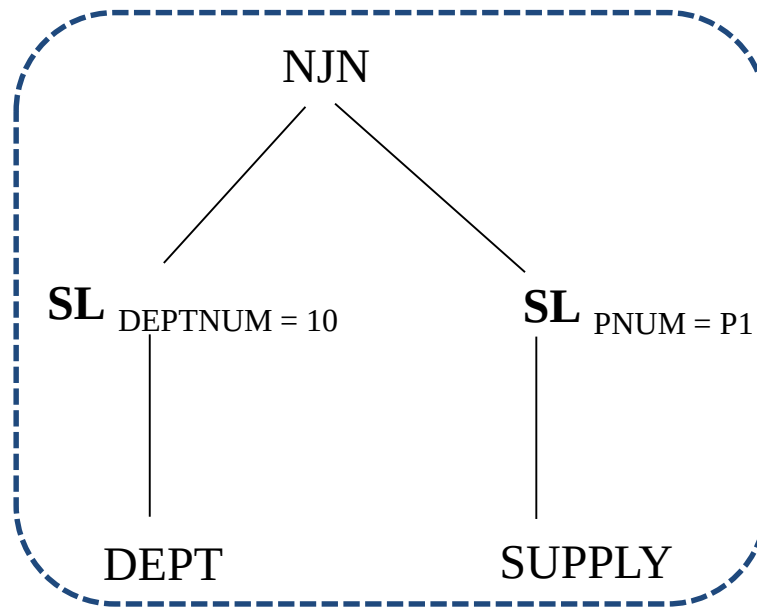
Finding Common Sub - Expression



Removing Common Sub - Expression



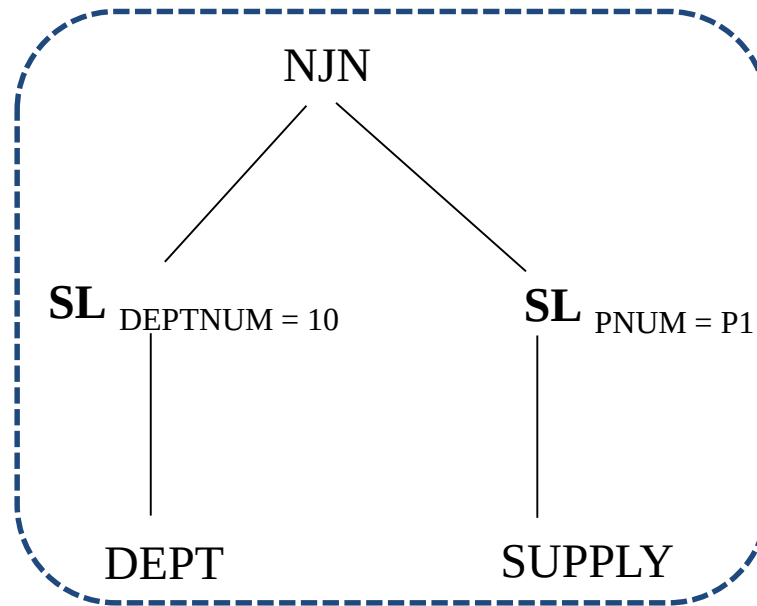
Removing Common Sub - Expression



Do we need to apply criteria 1 and/or 2? No, already simplified.

Transformed Query

Q_T: **SL**_{DEPTNUM = 10} *DEPT* **NJN** **SL**_{PNUM = P1} *SUPPLY*



Transformed Query

Output:

Q_T: **SL**_{DEPTNUM = 10} *DEPT* **NJN** **SL**_{PNUM = P1} *SUPPLY*

Input:

Q: (**SL**_{DEPTNUM = 10} *DEPT* **NJN** (**SL**_{PNUM = P1} *SUPPLY* **DF** **SL**_{PNUM = P2} *SUPPLY*))

UN (**SL**_{DEPTNUM = 10} *DEPT* **NJN** **SL**_{PNUM = P1} *SUPPLY*)

$$Q \longleftrightarrow Q_T$$

Last Example

EMP (EMPNUM, DEPTNUM, NAME, SAL, AGE)
DEPT (DEPTNUM, NAME, AREA, MGRNUM)

Consider the following global query:

$((SL_{F1} \text{ EMP } \Join_{A=B} \text{ DEPT}) \text{ UN } (SL_{F2} \text{ EMP } \Join_{A=B} \text{ DEPT})) \text{ DF } (SL_{F3} \text{ EMP } \Join_{A=B} \text{ DEPT})$

Here,

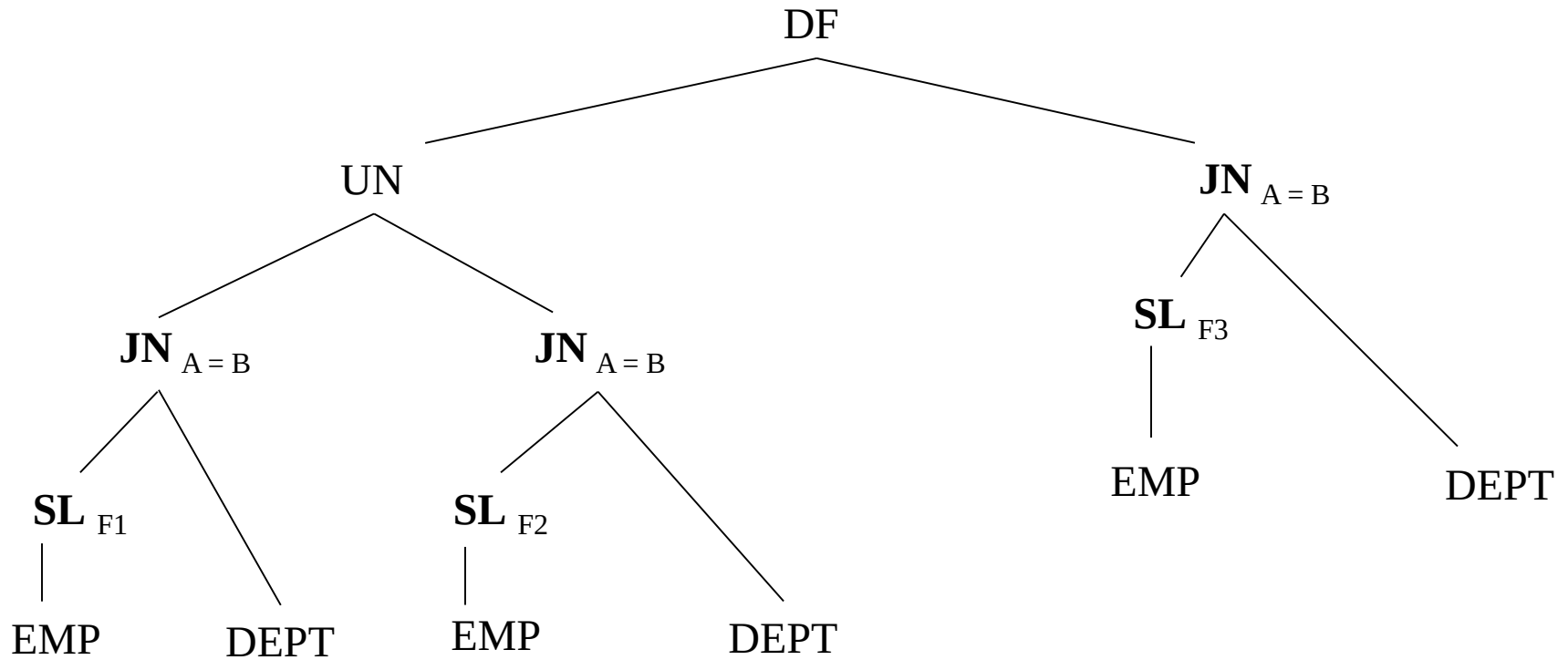
F1, F2, F3 can represent any condition. In this example consider none of them are same.

Imagine, $A = B = \text{DEPTNUM}$

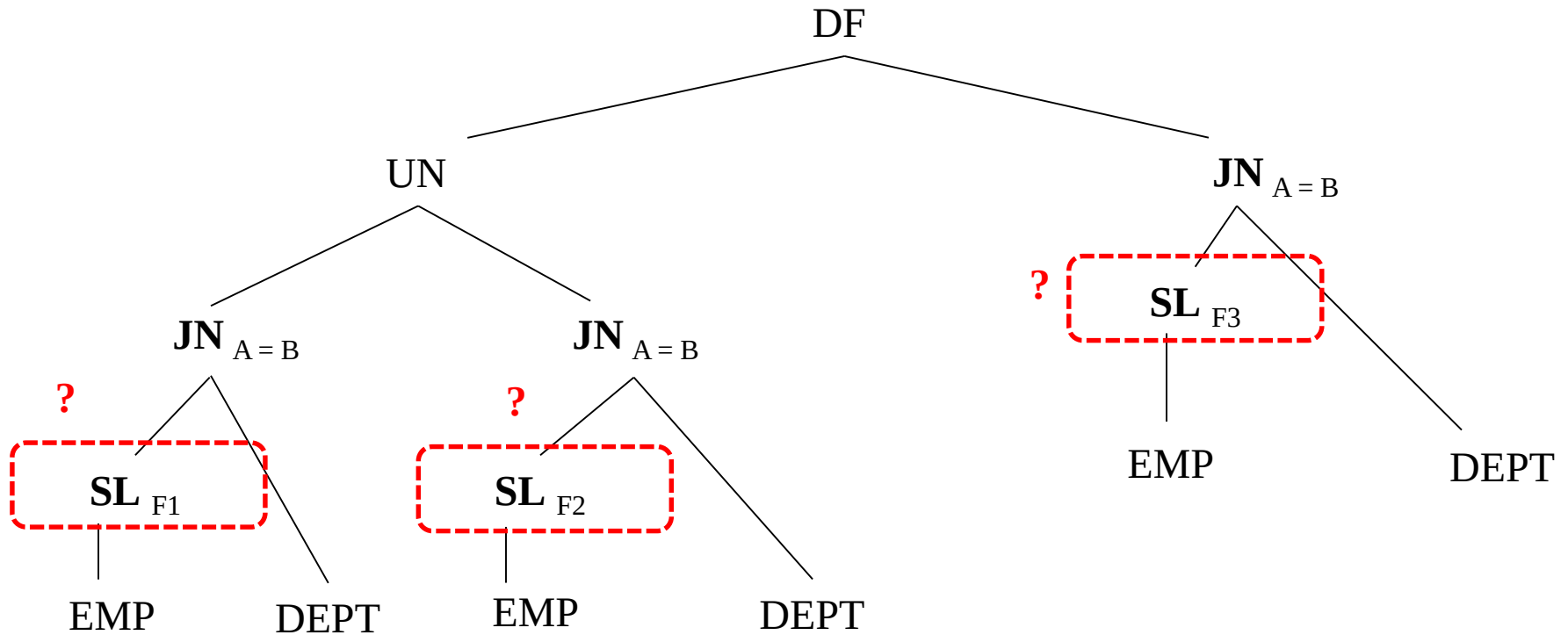
Now, answer the following questions.

- i. Draw the *operator tree*. [2]
- ii. Perform step-by-step transformations to simplify the operator tree, indicating which rule and criterion is applied at each step. [5]

Operator Tree

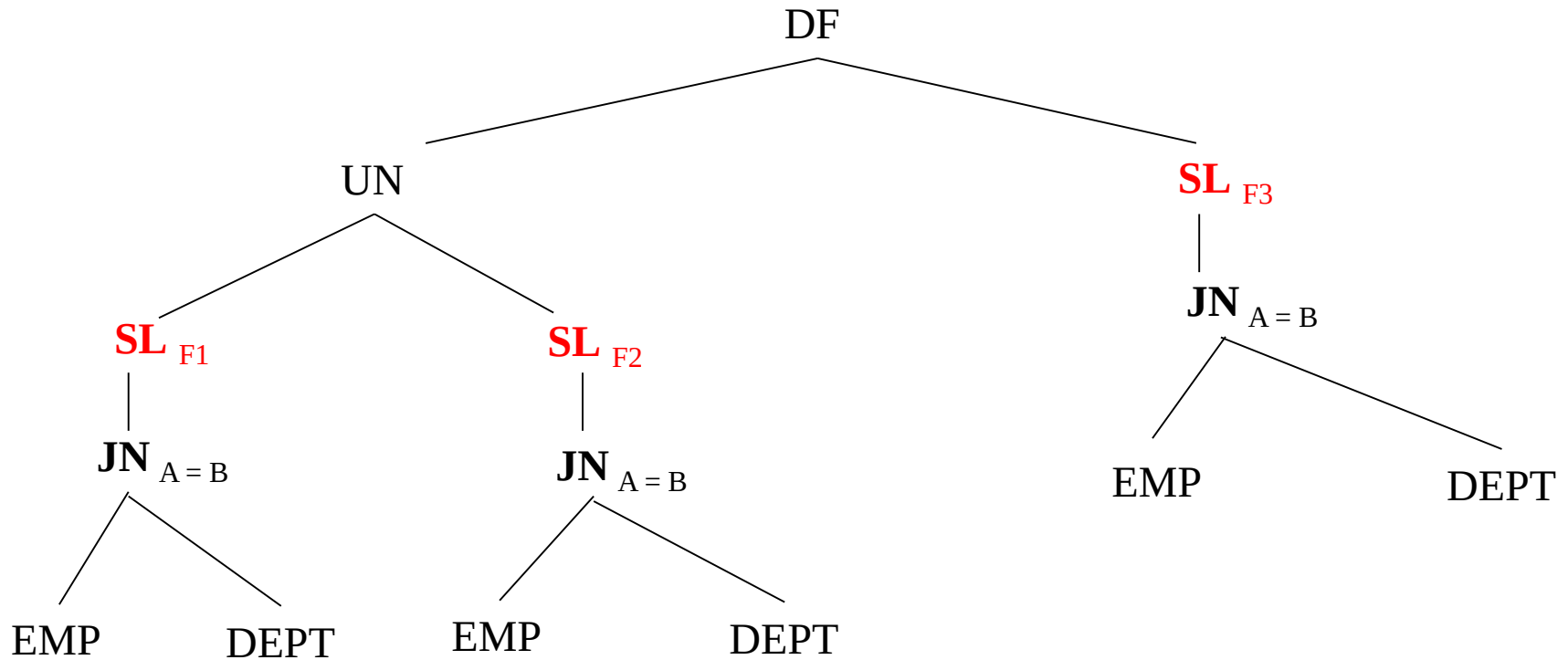


Finding Common Sub-Expression



Finding Common Sub-expression

Any common portion?

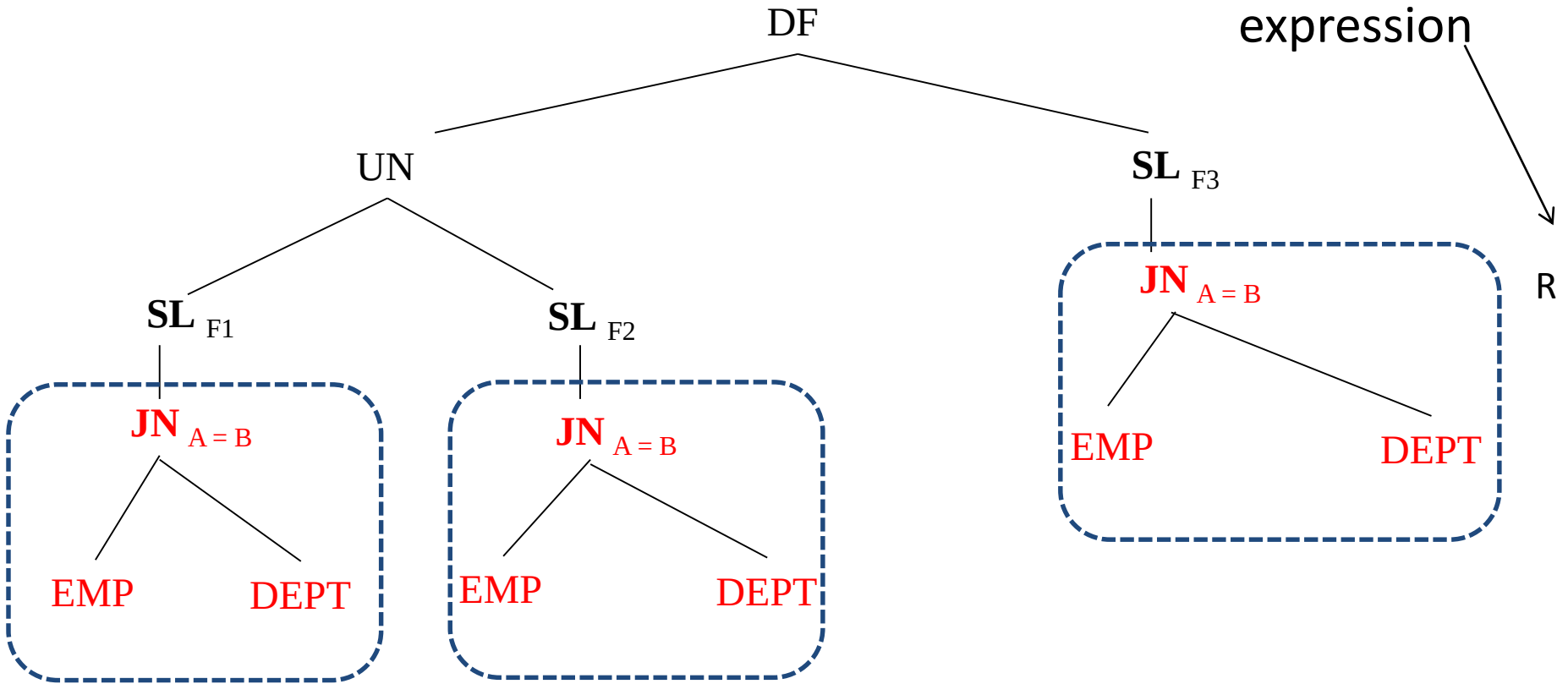


Finding Common Sub-expression

Any common portion?

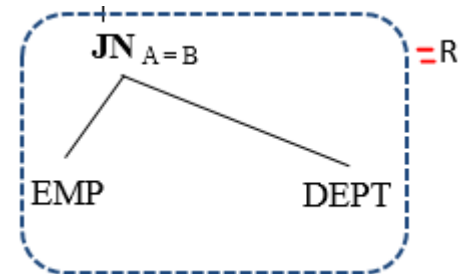
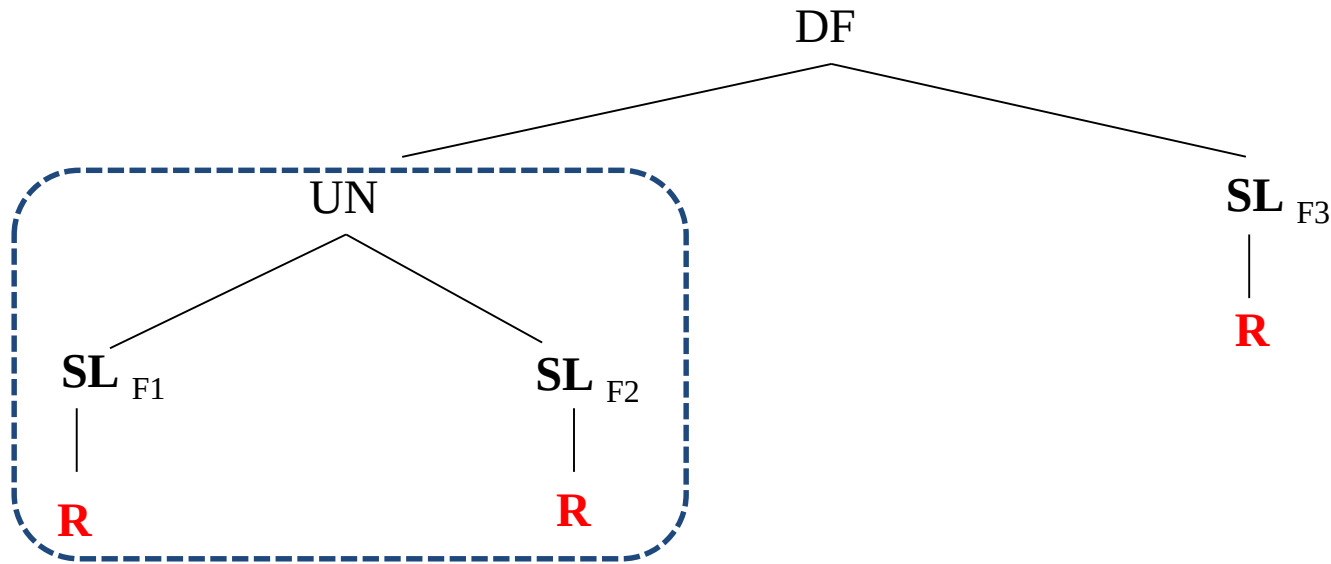
Common sub-expression

R



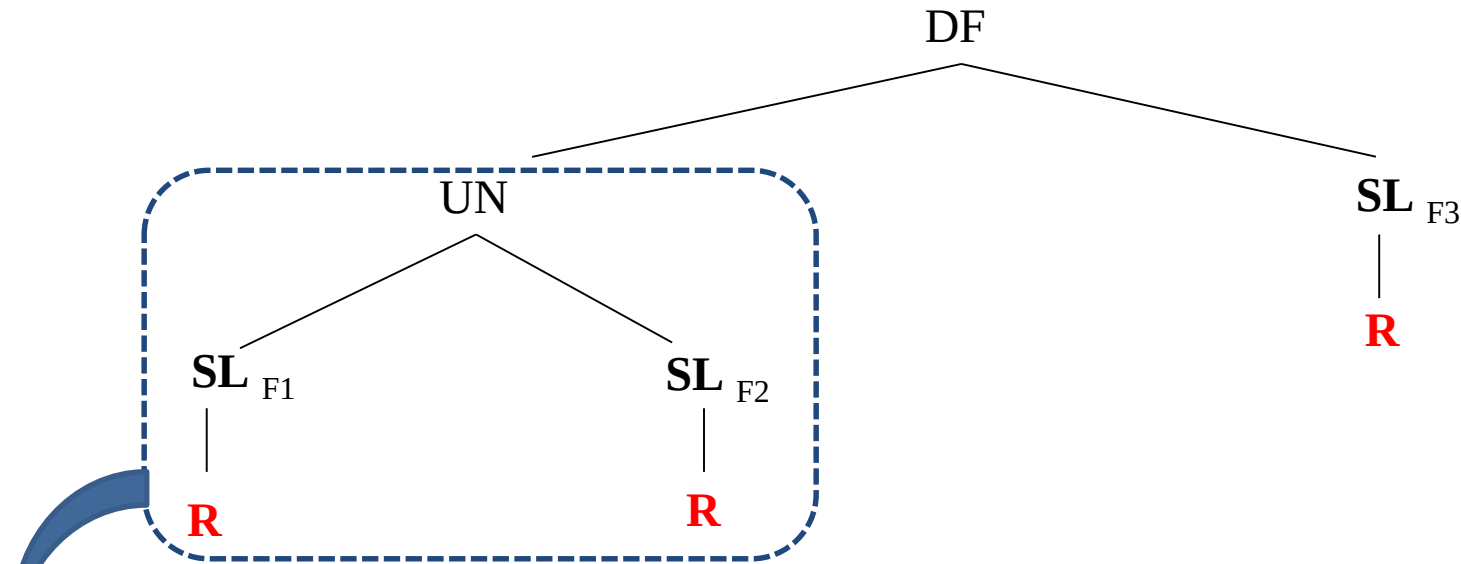
Finding Common Sub-expression

Any common portion?

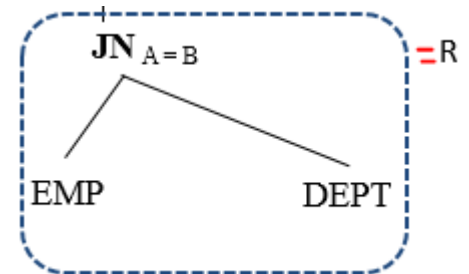


Finding Common Sub-expression

Any common portion?

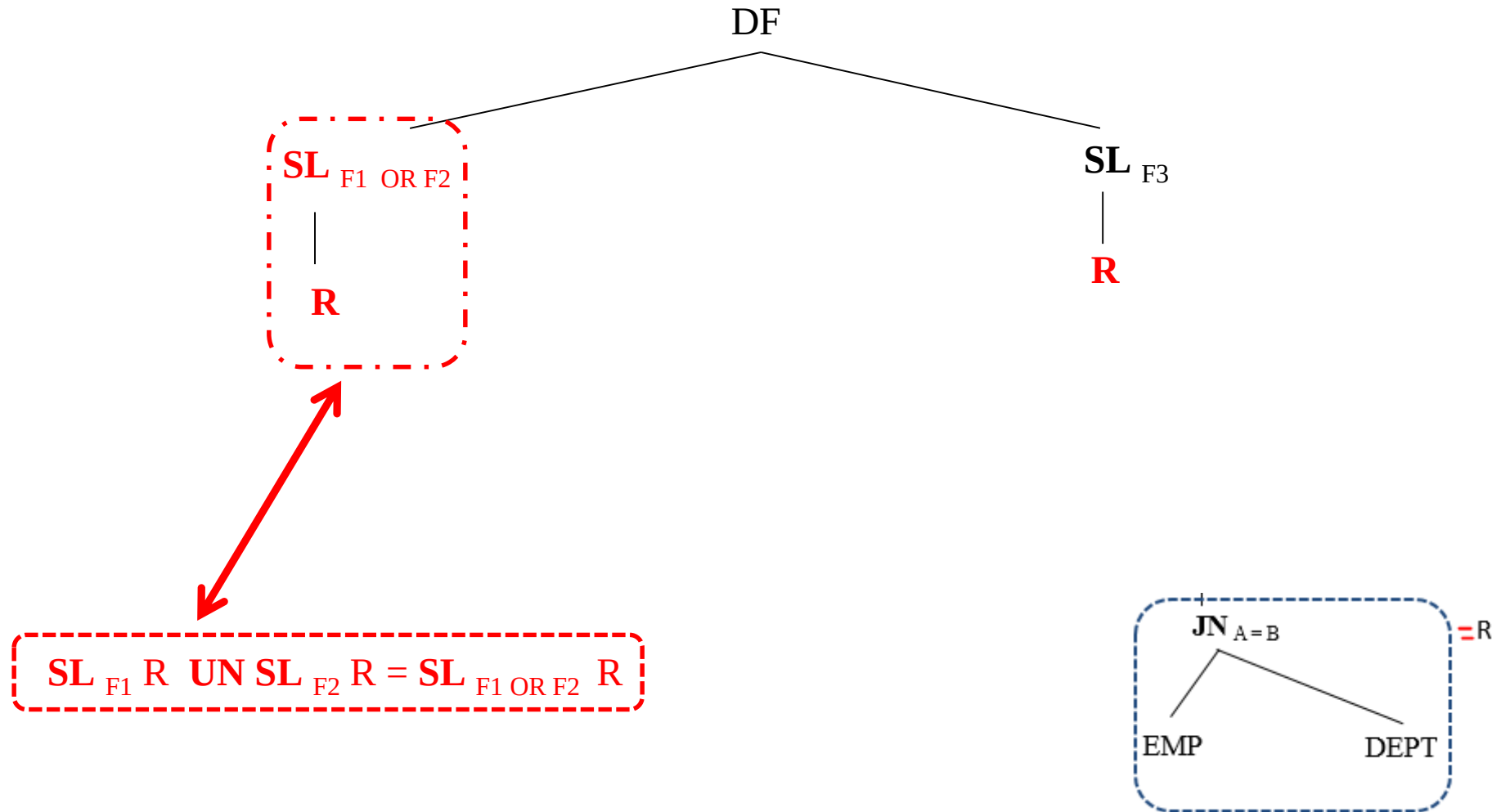


We can write it as $SL_{F1} R \text{ UN } SL_{F2} R$ which is Rule 8 !



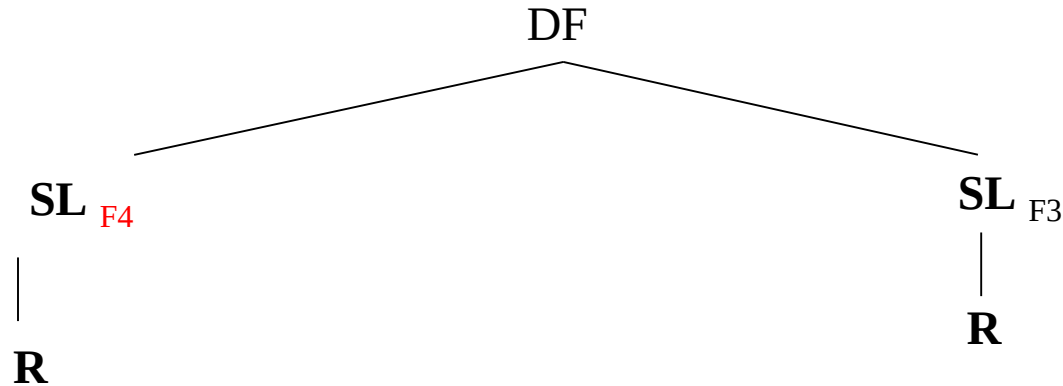
Removing Common Sub-expression

Any common portion?

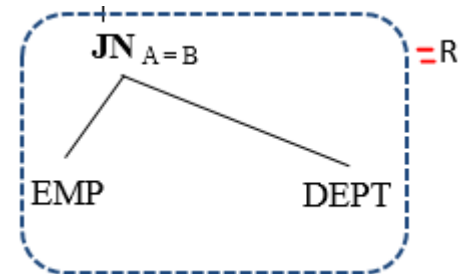


Finding Common Sub-expression

Any common portion?

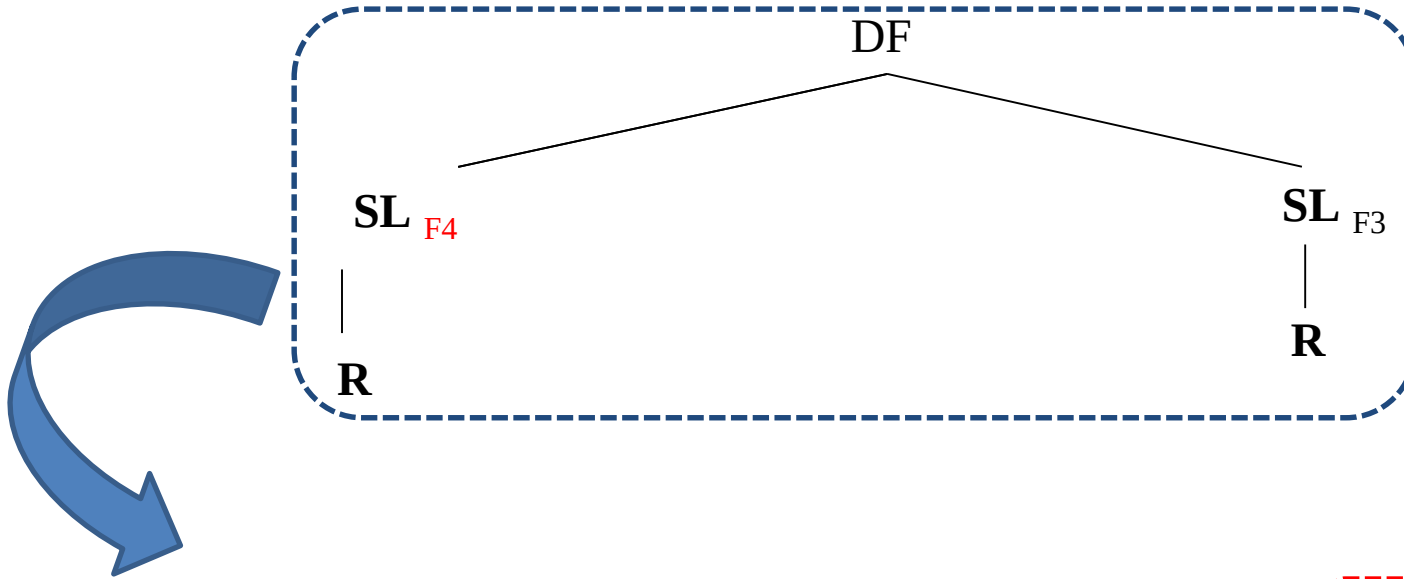


Let, $F4 = F1 \text{ OR } F2$



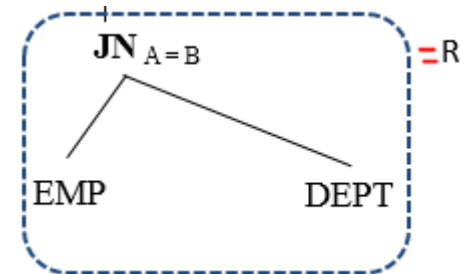
Finding Common Sub-expression

Any common portion?



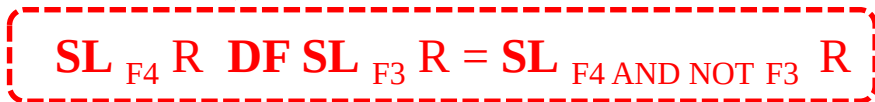
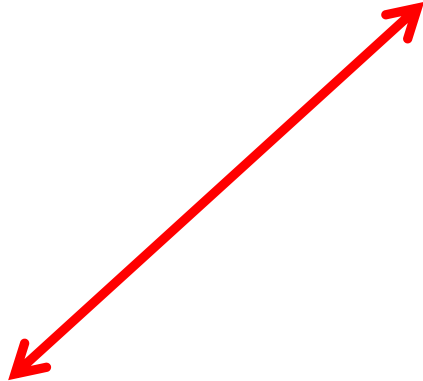
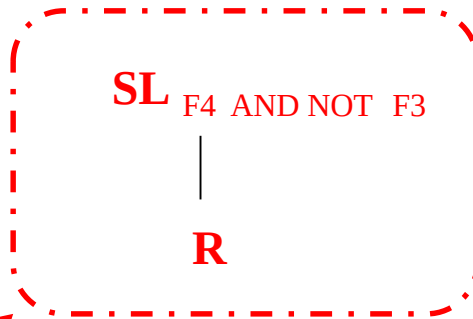
We can write it as **SL_{F4} R DF SL_{F3} R** which is Rule 9 !

Let, F4 = F1 OR F2

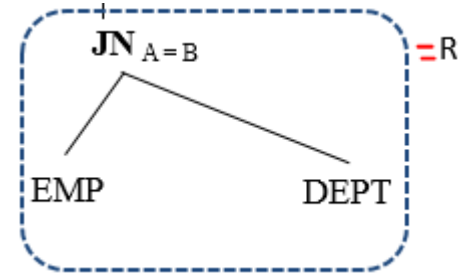


Removing Common Sub-expression

Any common portion?

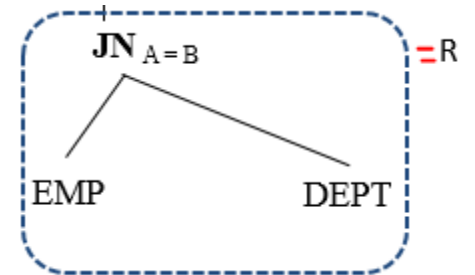


Let, $F4 = F1 \text{ OR } F2$

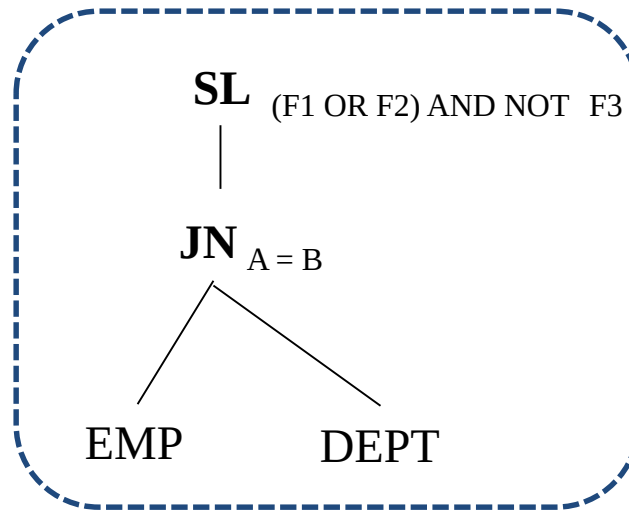


Removing Common Sub-expression

SL (F1 OR F2) AND NOT F3
|
R



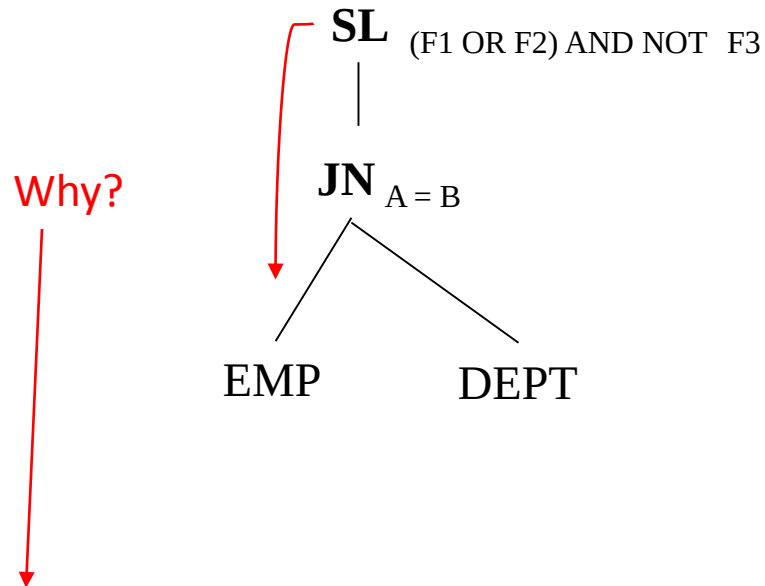
Removing Common Sub-expression



Can we apply Criterion 1 and/or 2?

Simplification

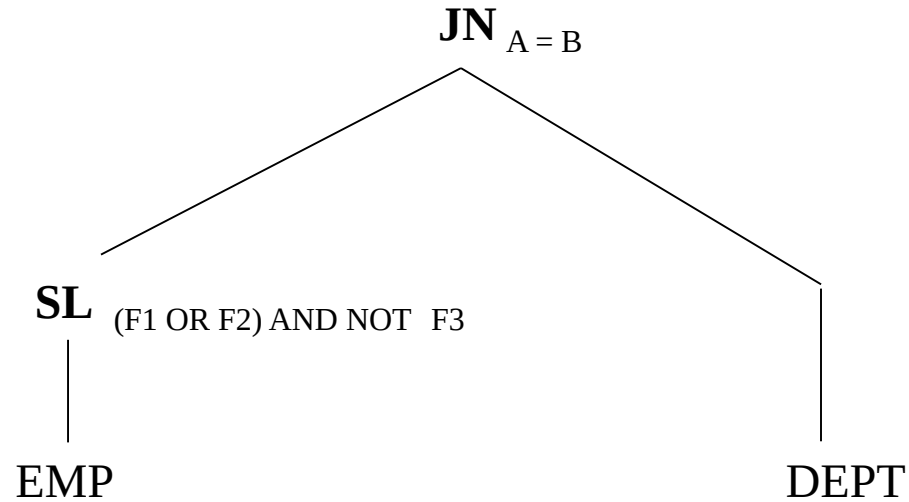
Applying Criterion 2 -



Because in the original query, all **SL**_{F1}, **SL**_{F2}, **SL**_{F3} were applied on EMP relation.

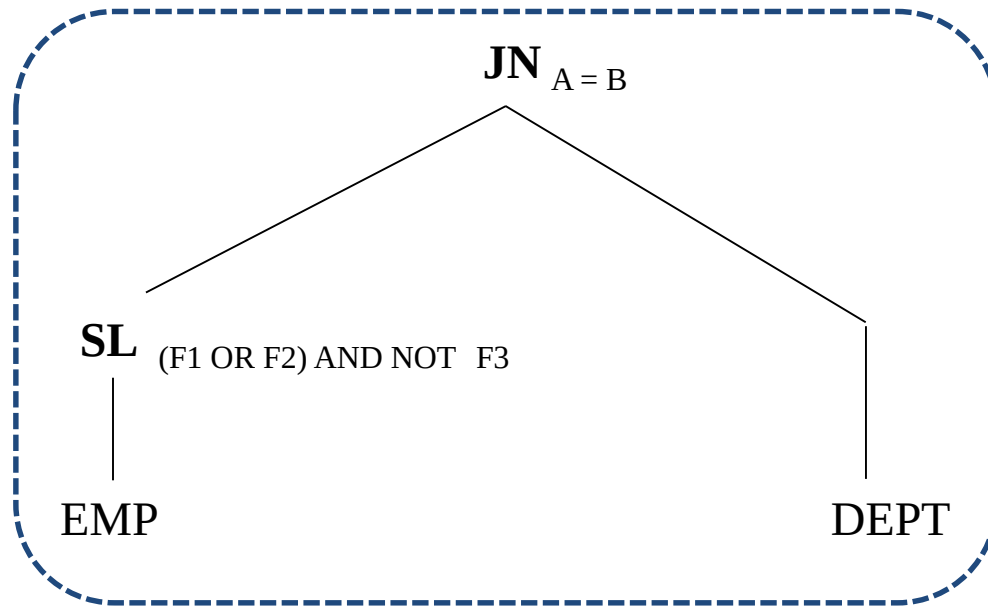
Simplification

After Applying Criterion 2 -



Transformed Query

Q_T: **SL**_{(F1 OR F2) AND NOT F3} *EMP* **JN**_{A = B} *DEPT*



Transformed Query

Output:

$Q_T: \text{SL}_{(F1 \text{ OR } F2) \text{ AND NOT } F3} \text{ EMP } \text{JN}_{A=B} \text{ DEPT}$

Input:

Q:

$((\text{SL}_{F1} \text{ EMP } \text{JN}_{A=B} \text{ DEPT}) \text{ UN } (\text{SL}_{F2} \text{ EMP } \text{JN}_{A=B} \text{ DEPT})) \text{ DF } (\text{SL}_{F3} \text{ EMP } \text{JN}_{A=B} \text{ DEPT})$

$$Q \longleftrightarrow Q_T$$

Exercise 1

② EMP (EMPNUM, DEPTNUM, NAME, SAL, AGE)
DEPT (DEPTNUM, NAME, AREA, MGRNUM)

Query: PJ NAME, AGE ((EMP JN DEPTNUM=DEPTNUM SL AREA="North"
DEPT) DF (EMP JN DEPTNUM=DEPTNUM SL DEPTNUM < 10 DEPT))

Exercise 2

4. Consider the following global relational schemata.

EMP (ID, NAME, SAL, AGE, MGRNUM, DEPTNUM)

DEPT (ID, AREA, DEPTNUM, MGRNUM)

Corresponding fragmentation schemata:

$EMP_1 = SL_{SAL \leq 25K} EMP$

$EMP_2 = SL_{SAL > 25K} EMP$

$DEPT_1 = SL_{AREA = "North"} DEPT$

$DEPT_2 = SL_{AREA = "South"} DEPT$

Also consider the following global query.

$PJ_{NAME, AREA}(((SL_{SAL > 25K} EMP \Join_{ID=ID} SL_{AREA = "North"} DEPT) \Join_{ID=ID} SL_{SAL \leq 25K} EMP \Join_{ID=ID} SL_{AREA = "North"} DEPT)) \Join_{ID=ID} (SL_{AREA = "North"} (EMP \Join_{ID=ID} DEPT)))$

Exercise 3

EMP (EMPNUM, DEPTNUM, NAME, SAL, AGE)
DEPT (DEPTNUM, NAME, AREA, MGRNUM)

Consider the following global query:

$$\left((SL_{F1} \text{ EMP } JN_{A=B} \text{ DEPT}) \text{ DF } (SL_{F2} \text{ EMP } JN_{A=B} \text{ DEPT}) \right) \text{ NJN } \\ \left((\text{EMP } JN_{A=B} \text{ DEPT}) \text{ UN } (SL_{F3} \text{ EMP } JN_{A=B} \text{ DEPT}) \right)$$

Here,
F1, F2, F3 can represent any condition. In this example consider none of them are same.
Imagine, $A = B = \text{DEPTNUM}$

Exercise 4

EMP (EMPNUM, DEPTNUM, NAME, SAL, AGE)
DEPT (DEPTNUM, NAME, AREA, MGRNUM)

Consider the following global query:

$$\left((SL_{F_1} \text{ EMP } JN_{A=B} \text{ DEPT}) \text{ UN } (SL_{F_2} \text{ EMP } JN_{A=B} \text{ DEPT}) \right) \text{ NJN } \\ \left((EMP \text{ } JN_{A=B} \text{ DEPT}) \text{ DF } (SL_{F_3} \text{ EMP } JN_{A=B} \text{ DEPT}) \right)$$

Here,
F1, F2, F3 can represent any condition. In this example consider none of them are same.
Imagine, $A = B = \text{DEPTNUM}$

Exercise 5

Consider the following global query and answer the questions from (i) to (iii).

$$\begin{aligned} &(((SL_{F1} R JN_{A=B} S) DF (SL_{F2} R JN_{A=B} S)) NJN (R JN_{A=B} S)) \\ &UN (SL_{F1 AND NOT F2} R JN_{A=B} S) \end{aligned}$$

- i) Draw the operator tree. [2]
- ii) Perform step-by-step transformations to simplify the operator tree, indicating which rule and criterion is applied at each step. [5]
- iii) Write the query from the obtained simplified tree. [2]

Here,
F1, F2, F3 can represent any condition. In this example consider none of them are same.
Imagine, A and B represents the same attribute of two different relation R and S.

Exercise 6

EMP (*EMPNUM*, *DEPTNUM*, *NAME*, *SAL*, *AGE*)
DEPT (*DEPTNUM*, *NAME*, *AREA*, *MGRNUM*)

(a) Consider the following global query and answer the questions (i) and (ii).

$$\left(\left((SL_{SAL} > 25K \text{ EMP } \Join_{ID=ID} \text{ DEPT}) \text{ DF } (SL_{AGE} \leq 25 \text{ EMP } \Join_{ID=ID} \text{ DEPT}) \right) \text{ NJN } (\text{EMP } \Join_{ID=ID} \text{ DEPT}) \right)$$
$$\text{DF } (SL_{SAL} > 25K \text{ AND } AGE > 25 \text{ EMP } \Join_{ID=ID} \text{ DEPT})$$

- i. Draw the operator tree. [2]
- ii. Perform step-by-step transformations to simplify the operator tree, indicating which rule and criterion is applied at each step. [6]